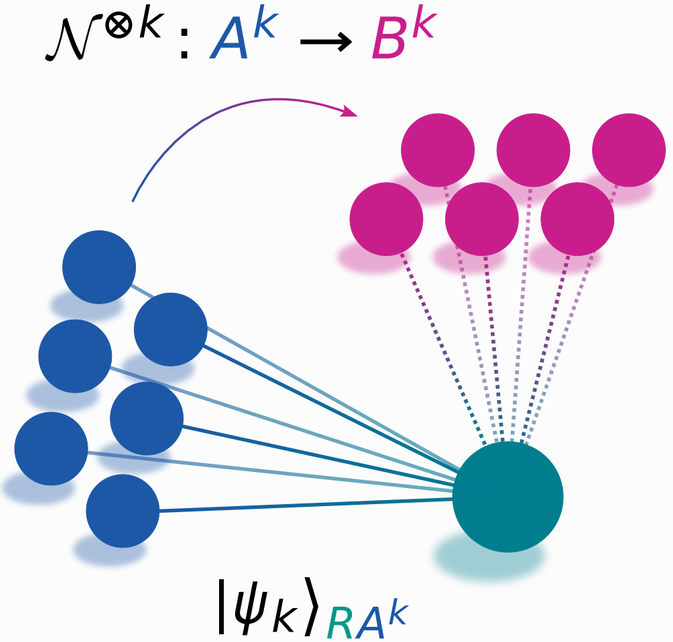
Multipartite entanglement, noise, and quantum information processing

arXiv:2508.09978

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Department of Mathematics and IQUIST
University of Illinois Urbana-Champaign





Joint work with my student
Sujeet Bhalerao (UIUC Math)

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

NSF

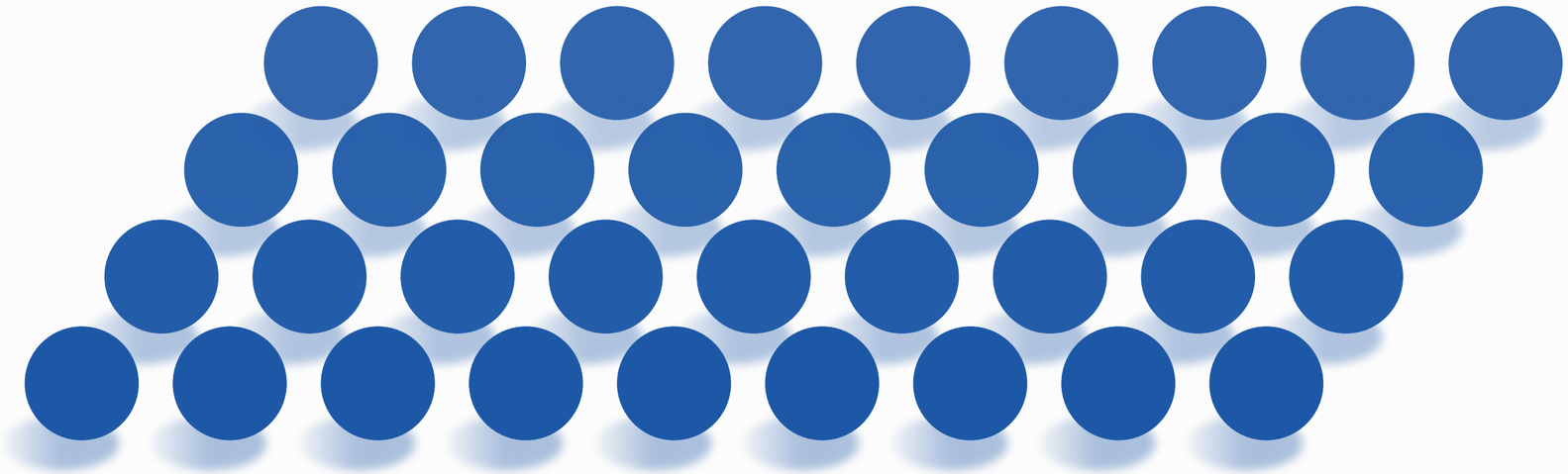
Illinois Quantum Information Science and Technology Center

Grainger College of Engineering
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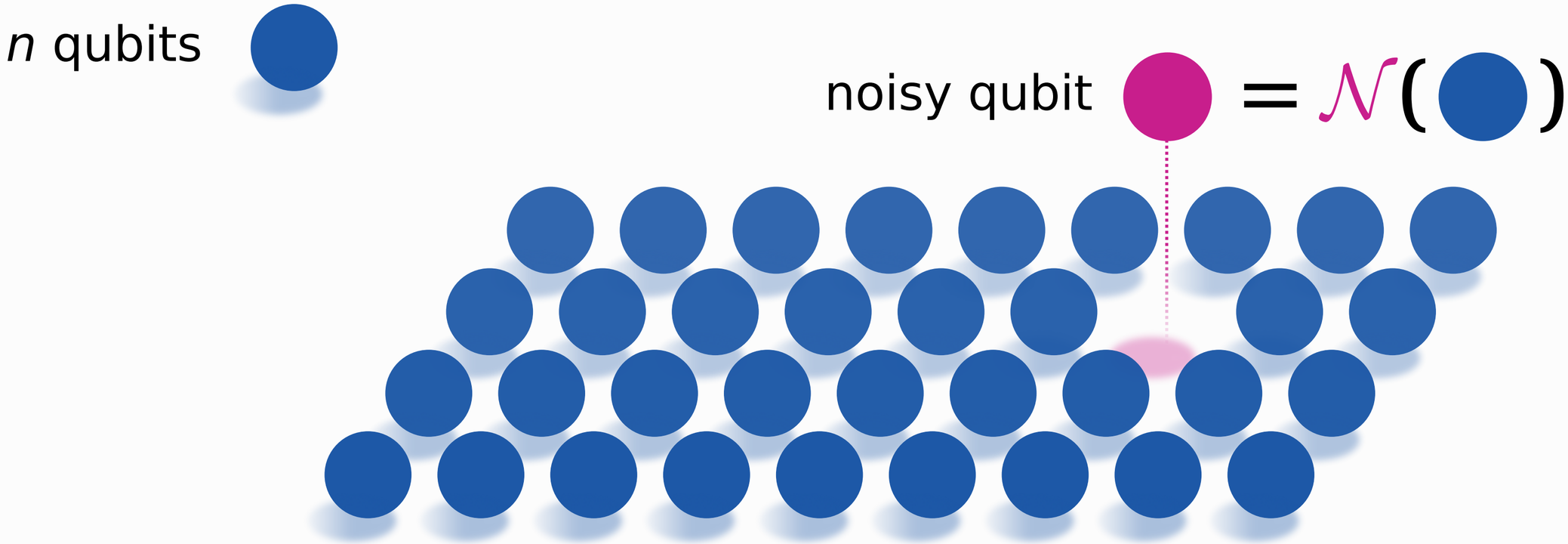
Noise in quantum devices

n qubits



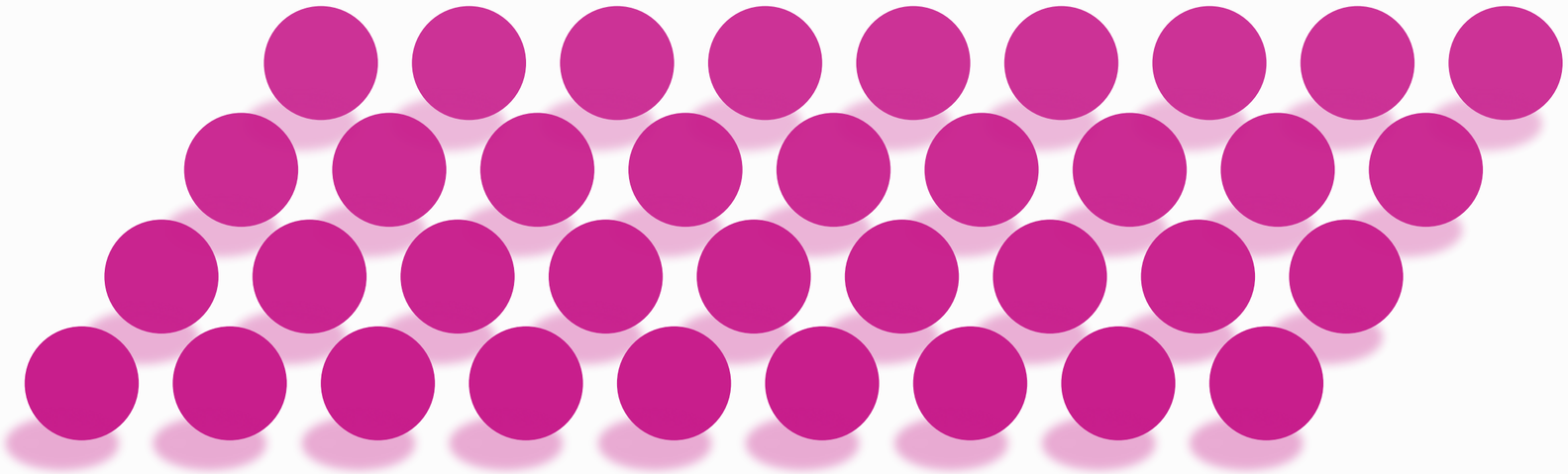


Noise in quantum devices



Noise in quantum devices

n noisy qubits  $= \mathcal{N}($ $)$



Independent and identically distributed (i.i.d.) noise model $\mathcal{N}^{\otimes n}$

Talk outline

- Quantum information 101
- Coherent information and quantum capacity
- Symmetries in quantum information theory
- Efficient algorithms to compute entropies of symmetric states
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Quantum information 101: Pure states

Qubit  : Quantum-mechanical two-level system

Examples: Polarization of a photon, spin alignment of an electron, ...

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State space:

$$0 \longmapsto |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$1 \longmapsto |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

General “pure” state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{C}^2$$

where $|\alpha|^2 + |\beta|^2 = 1$.

Quantum information 101: Mixed states

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$$\rho = \sum_i p_i |\psi\rangle\langle\psi|$$

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If ρ and σ are two **commuting** density operators ($\rho\sigma = \sigma\rho$),

they can be diagonalized simultaneously,

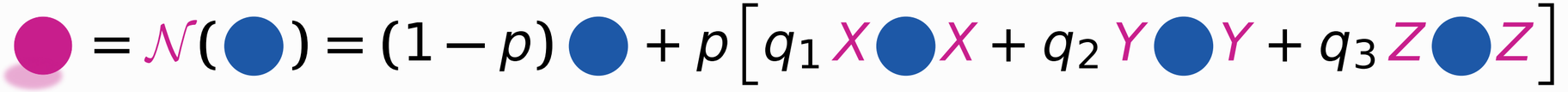
$$\rho = \begin{pmatrix} \lambda & 0 \\ 0 & 1-\lambda \end{pmatrix} \text{ and } \sigma = \begin{pmatrix} \mu & 0 \\ 0 & 1-\mu \end{pmatrix},$$

and the system in one of these states behaves essentially **classically**.

Quantum Info 101: Quantum channels

Noise in quantum systems is modeled by a **quantum channel**.

Pauli channels: simple but physically relevant.

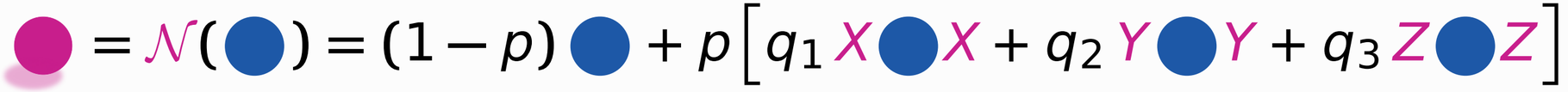


for $p \in [0, 1]$ and a probability distribution (q_1, q_2, q_3), and

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for $p \in [0, 1]$ and a probability distribution (q_1, q_2, q_3), and

bit flip $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ bit-phase flip $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ phase flip $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$X|0\rangle = |1\rangle \qquad Y|0\rangle = i|1\rangle \qquad Z|0\rangle = |0\rangle$$
$$X|1\rangle = |0\rangle \qquad Y|1\rangle = -i|0\rangle \qquad Z|1\rangle = -|1\rangle$$

Quantum information 101: Entanglement

Two qubits  have state space $\mathbb{C}^2 \otimes \mathbb{C}^2$.

$$\mathbb{C}^2 \otimes \mathbb{C}^2 \ni |\psi\rangle = \alpha|0\rangle \otimes |0\rangle + \beta|0\rangle \otimes |1\rangle + \gamma|1\rangle \otimes |0\rangle + \delta|1\rangle \otimes |1\rangle.$$

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Mathematical fact: Not every vector in $\mathbb{C}^2 \otimes \mathbb{C}^2$ is of the form $|\phi_1\rangle \otimes |\phi_2\rangle$.

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Physical consequence: Entanglement!

"EPR state" or "Bell pair" $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$



Quantum information 101: Entanglement

“EPR state” or “Bell pair” $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$ can be used for:

- Quantum teleportation (quantum communication)
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Bipartite pure-state entanglement characterized by mixedness of marginal:

"The more mixed the marginal, the more entangled the state."

Quantum information 101: Entanglement

"The more mixed the marginal, the more entangled the state."

EPR state $|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$ is maximally entangled:

$$\mathrm{tr}_A\, \Phi^+_{AB} = \tfrac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \mathrm{tr}_B\, \Phi^+_{AB}$$

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Product state $|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\omega\rangle_B$ is unentangled:

$$\mathrm{tr}_A\, \psi_{AB} = |\omega\rangle\langle\omega|_B \text{ and } \mathrm{tr}_B\, \psi_{AB} = |\phi\rangle\langle\phi|_A$$

Quantum information 101: Entanglement

"Mixedness" can be expressed via the spectrum of the marginal:

$\mathrm{tr}_A\, \Phi^+_{AB} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \mathrm{tr}_B\, \Phi^+_{AB}$: Spectrum: $(\lambda_1, \lambda_2) = (1/2, 1/2)$.

$\mathrm{tr}_A[\phi_A \otimes \omega_B] = |\omega\rangle\langle\omega|_B$ and $\mathrm{tr}_B[\phi_A \otimes \omega_B] = |\phi\rangle\langle\phi|_A$: Spectrum: $(\lambda_1, \lambda_2) = (1, 0)$.

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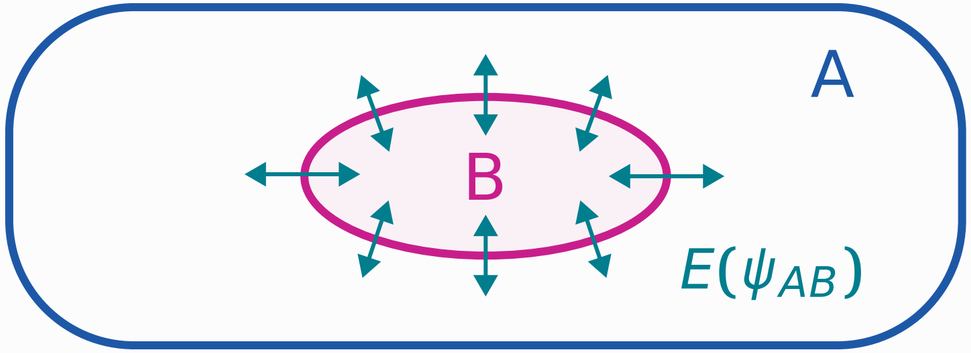
$\mathrm{tr}_A\, \Phi_{AB}^+ = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \mathrm{tr}_B\, \Phi_{AB}^+$: Spectrum: $(\lambda_1, \lambda_2) = (1/2, 1/2)$.

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Entanglement entropy

$$E(\psi_{AB}) = S(\mathrm{tr}_A\, \psi_{AB}) = -\mathrm{tr}\, \psi_B \log \psi_B$$
$$= -\lambda_1 \log \lambda_1 - \lambda_2 \log \lambda_2$$

$$E(\phi_A \otimes \omega_B) = 0 \leq E(\psi_{AB}) \leq 1 = E(\Phi_{AB}^+)$$



Quantum info 101: Multipartite entanglement

Three or more quantum systems can be entangled in much more intricate and complicated ways.

Ex.: three qubits can be entangled in two different ways. [Dür, Cirac, Vidal '00]

$$|\mathrm{GHZ}_3\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) \qquad |W_3\rangle = \frac{1}{\sqrt{3}}(|100\rangle + |010\rangle + |001\rangle)$$

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Multipartite entanglement is hard to describe!

Current state of entanglement theory

	pure	**mixed**
bipartite		
multipartite		

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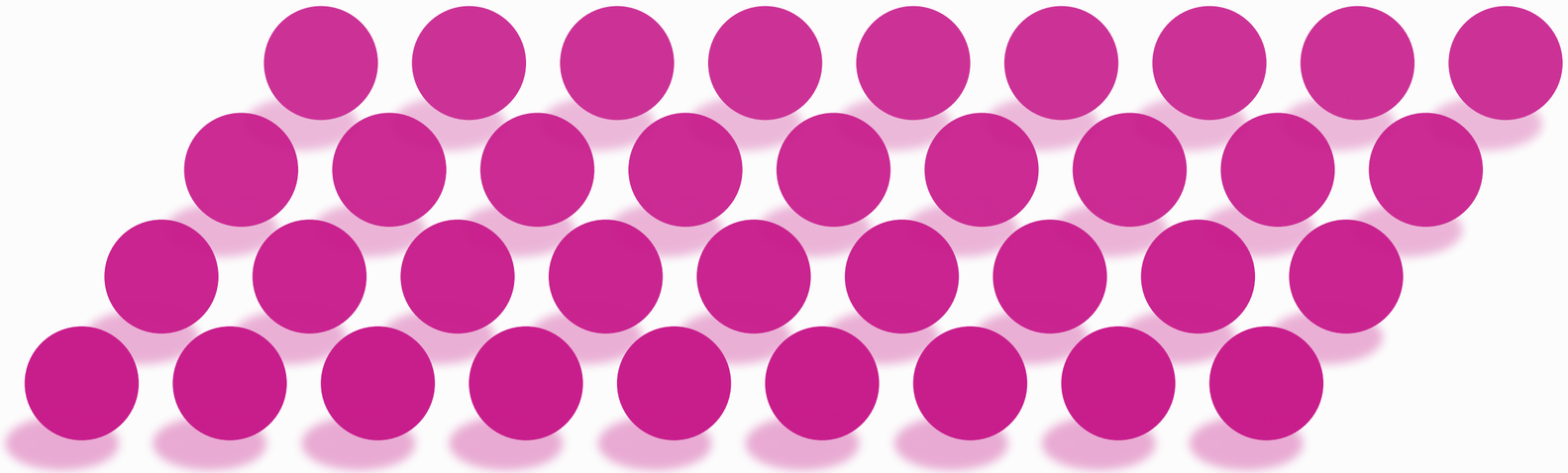
this talk!

Talk outline

- Quantum information 101
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Noise in quantum devices

n noisy qubits $\rho = \mathcal{N}(\rho) = (1-p)\rho + p\left[q_1 X\rho X + q_2 Y\rho Y + q_3 Z\rho Z\right]$



Independent and identically distributed (i.i.d.) noise model $\mathcal{N}^{\otimes n}$

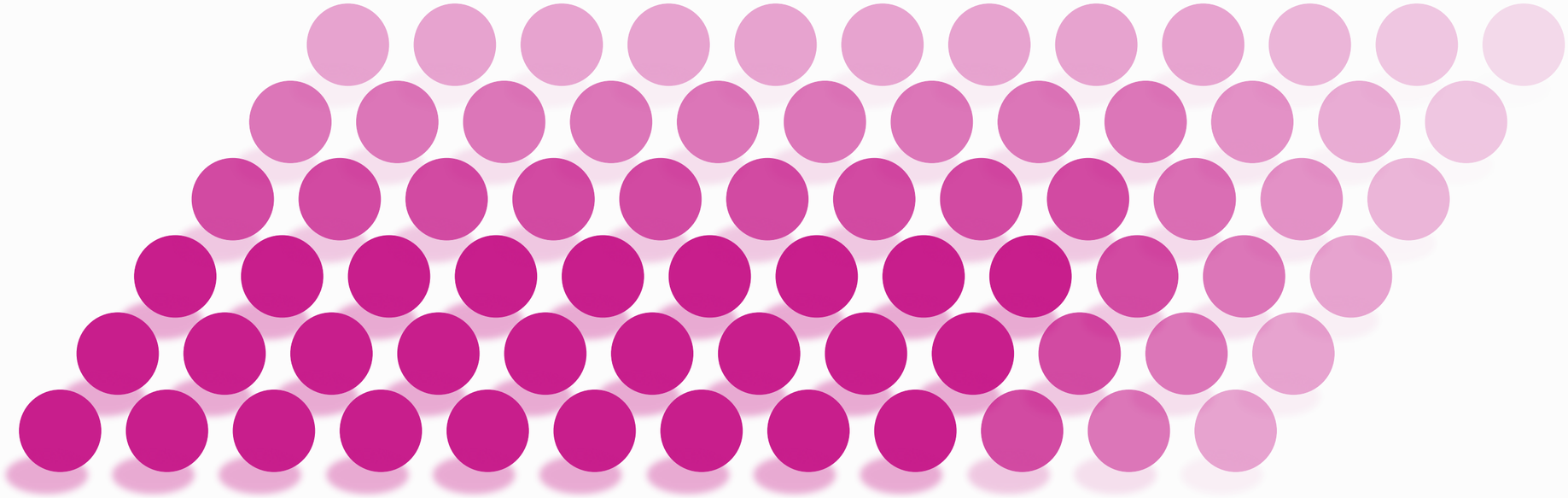
Noise in quantum devices

How much quantum information can be stored in a device  ?

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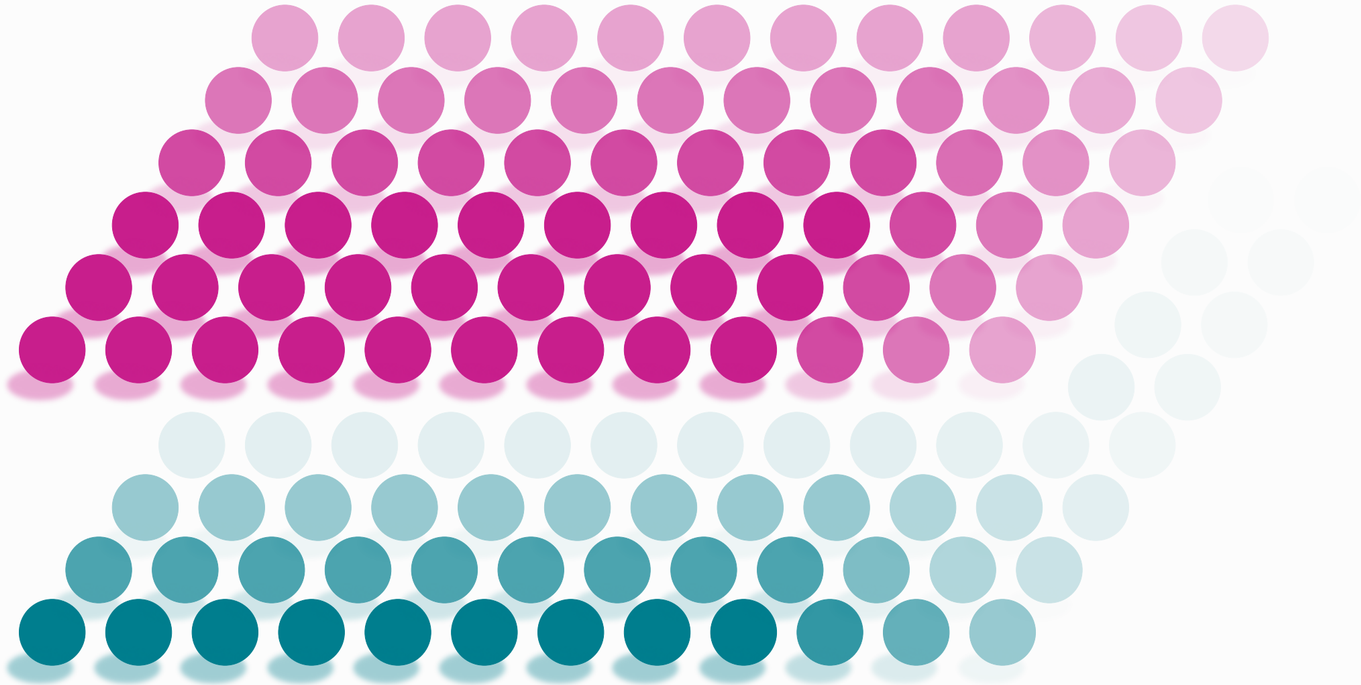
Step 1: Extend n-qubit system infinitely ($n \to \infty$).



Noise in quantum devices

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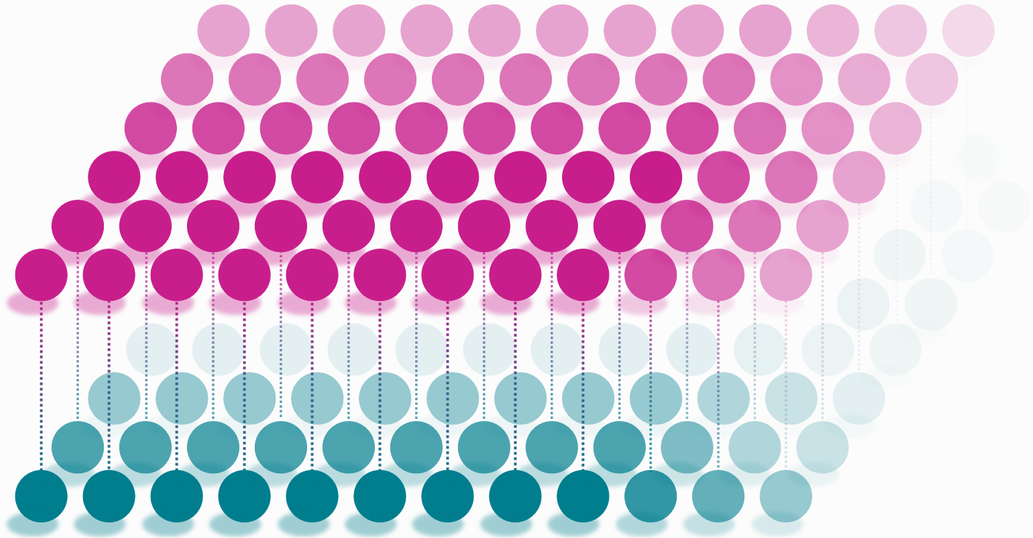
Step 2: Consider a **reference qubit** for each **system qubit**.

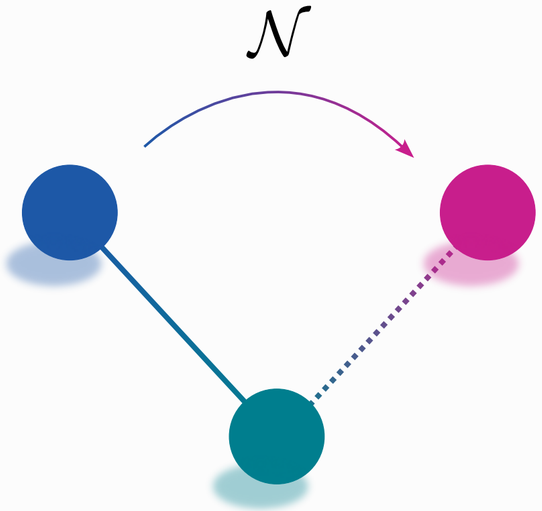


Noise in quantum devices

How much quantum information can be stored in a device  ?

Step 3: Quantify how much initial correlation between **system** and **reference qubits** survives the noise.

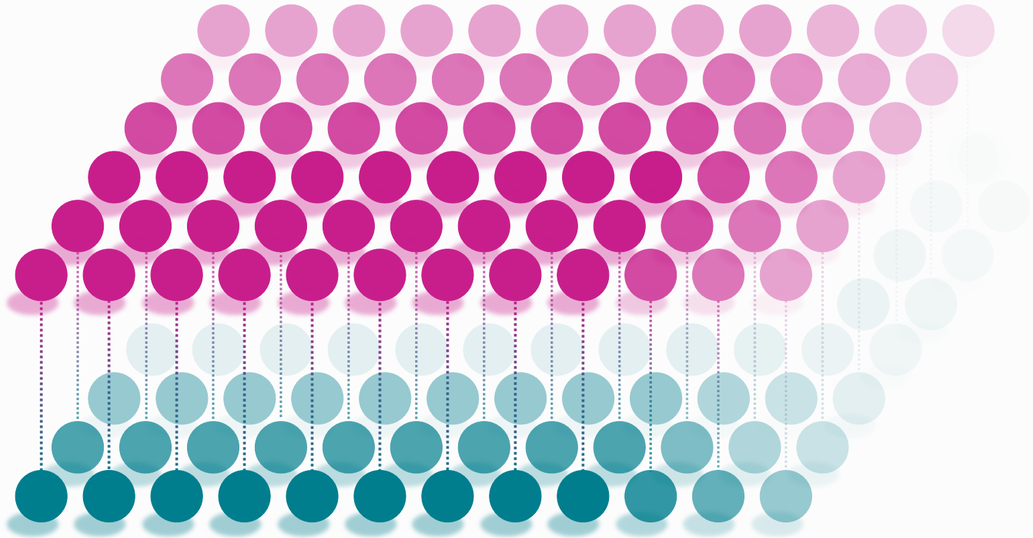


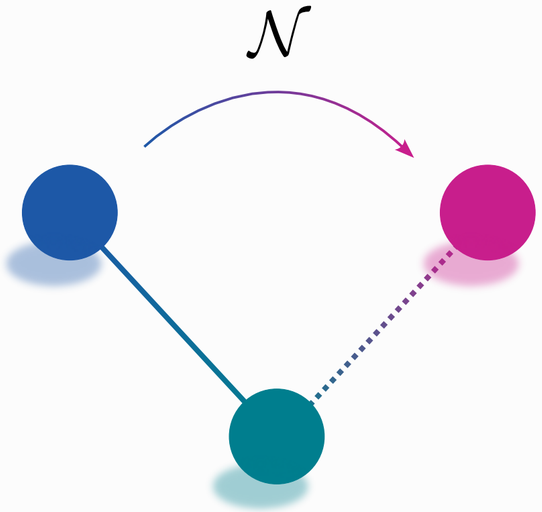


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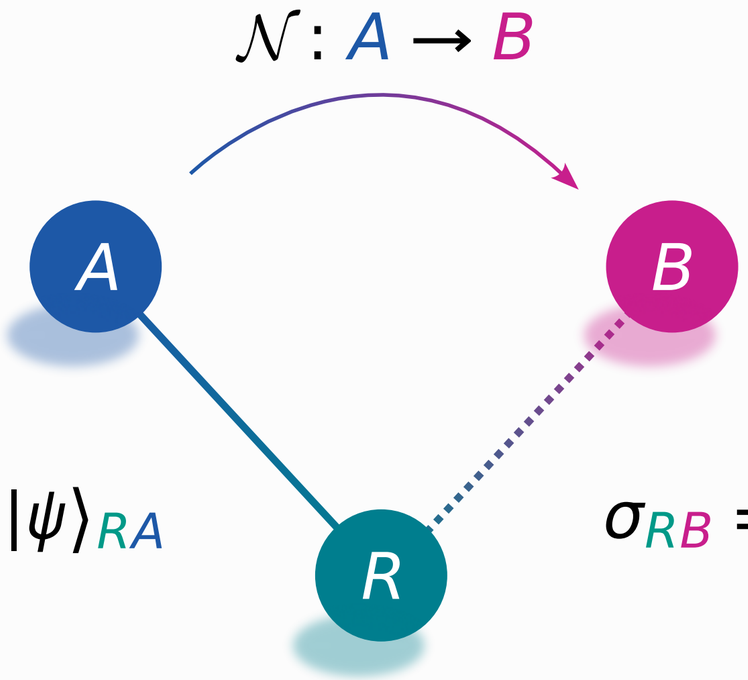
Step 3: Quantify how much initial correlation between **system** and **reference qubits** survives the noise.





How to quantify correlations?

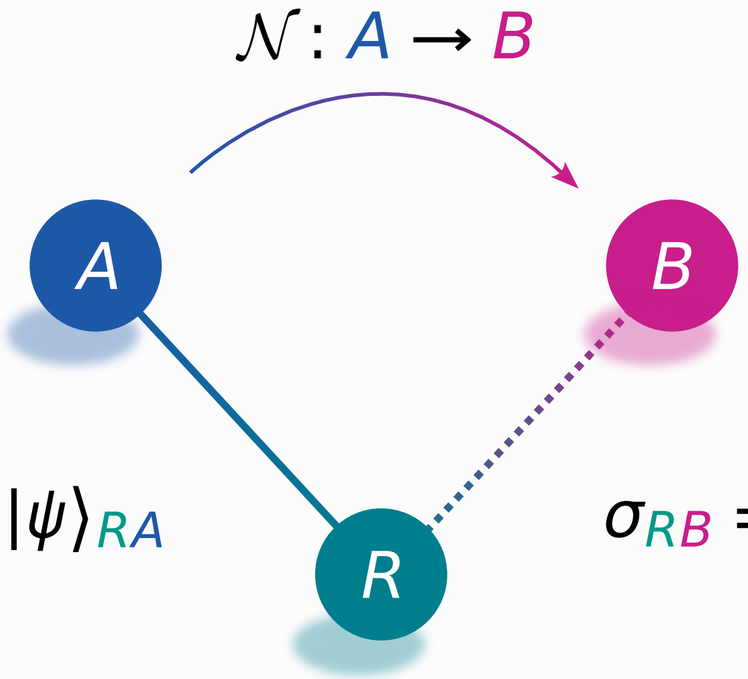
Quantifying quantum correlations



$|\psi\rangle_{RA}$

$\sigma_{RB} = (\mathrm{id}_R \otimes \mathcal{N}_{A\to B})(\psi_{RA})$

Quantifying quantum correlations



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Coherent information

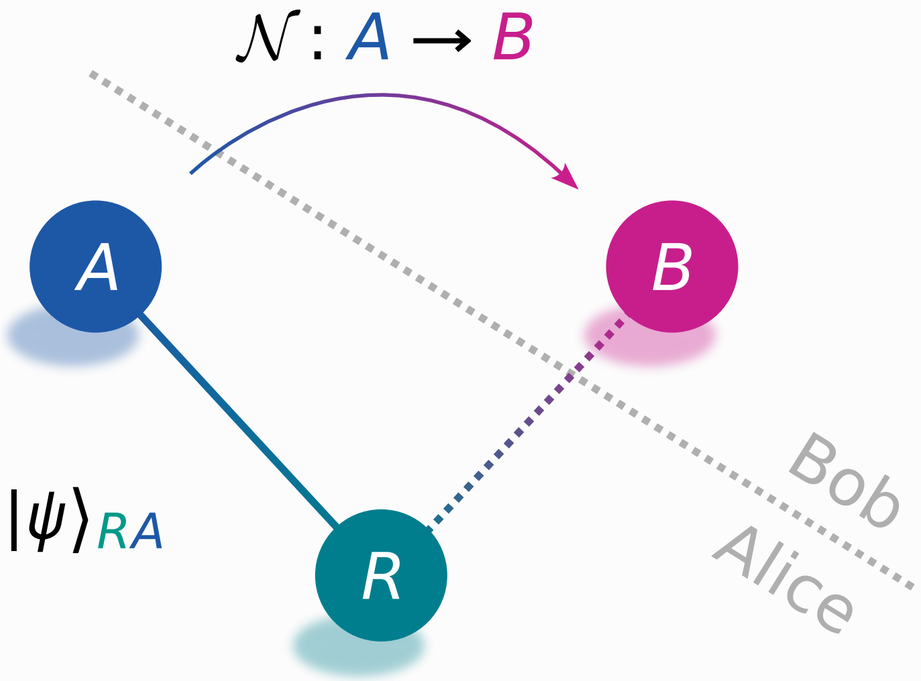
$$I_c(\mathcal{N}, \psi) = S(B)_\sigma - S(RB)_\sigma$$

$$S(Q)_\sigma = -\mathrm{tr}\left(\sigma_Q \log \sigma_Q\right)$$
$$= -\sum_i \lambda_i \log \lambda_i$$

where λ_i are the eig.-val's of σ.

Quantifying quantum correlations

Coherent information $I_c(\mathcal{N}, \psi) = S(B)_\sigma - S(RB)_\sigma$



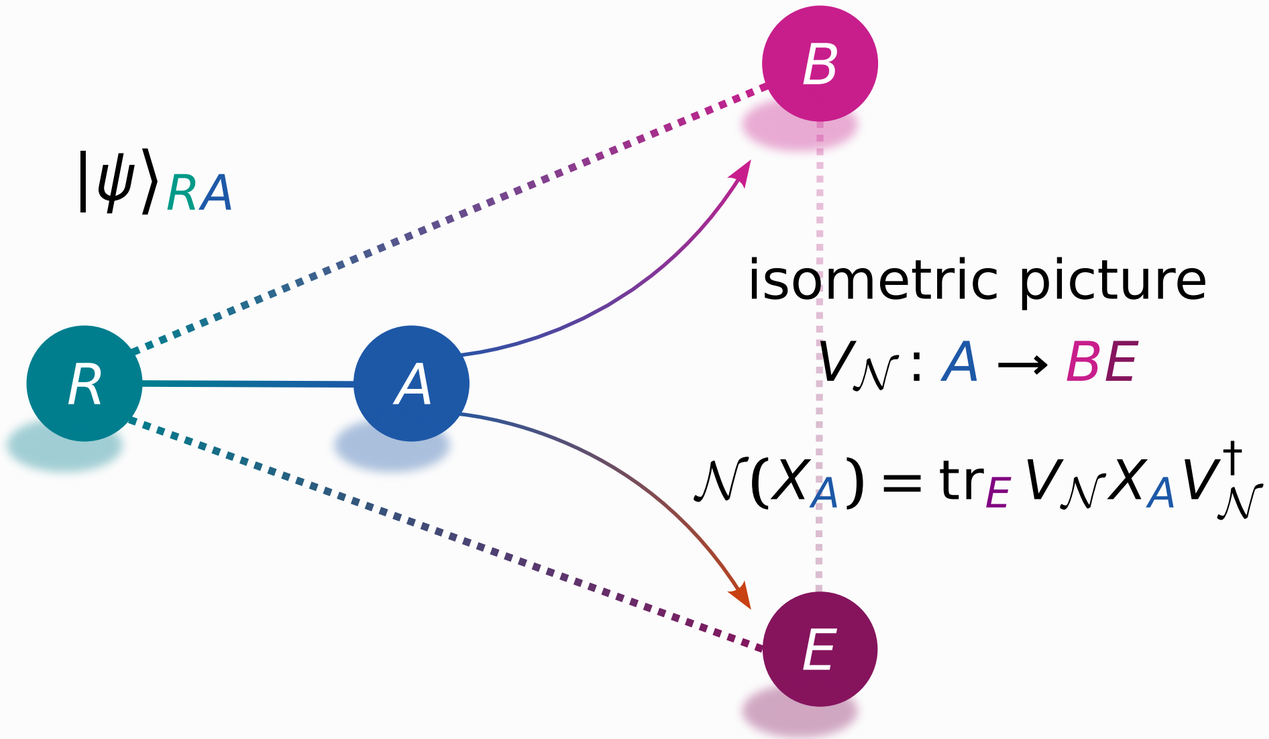
Interpretation of $I_c(\mathcal{N}, \psi)$:

Correlations established between **Alice** and **Bob** that survived the noisy channel action.

Quantifying quantum correlations

Coherent information

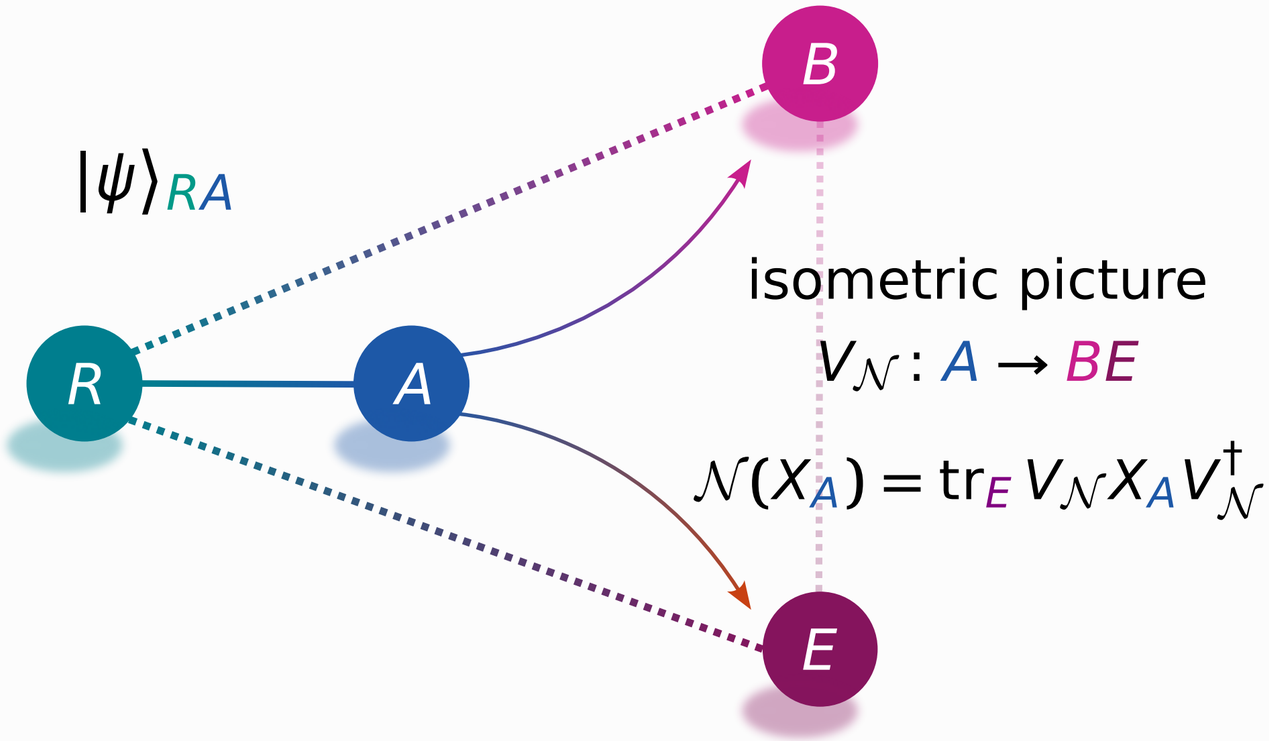
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Quantifying quantum correlations

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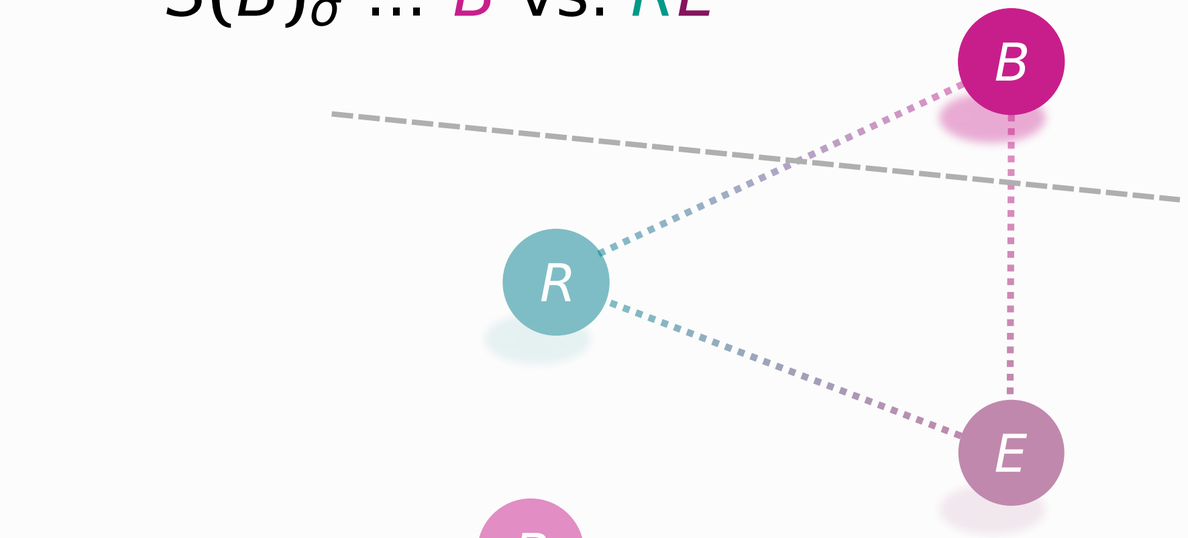


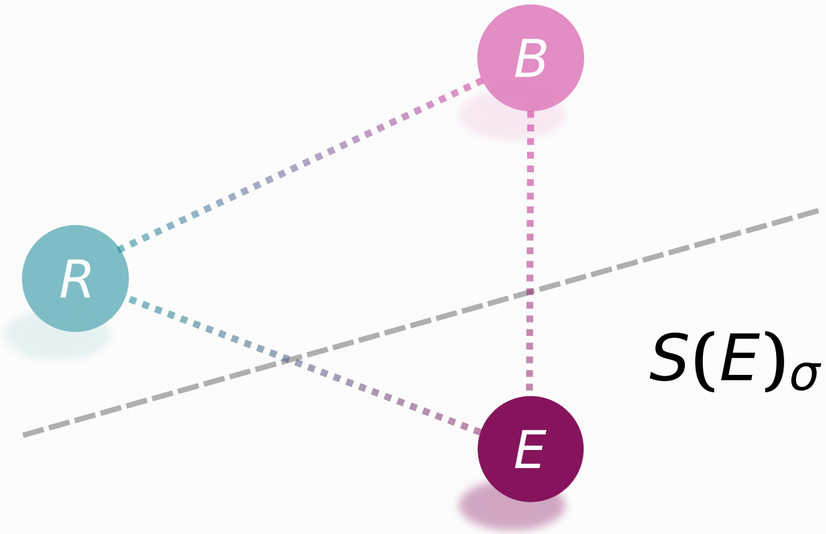
isometric picture

$V_{\mathcal{N}} : A \to BE$

$\mathcal{N}(X_A) = \mathrm{tr}_E \, V_{\mathcal{N}} X_A V_{\mathcal{N}}^\dagger$

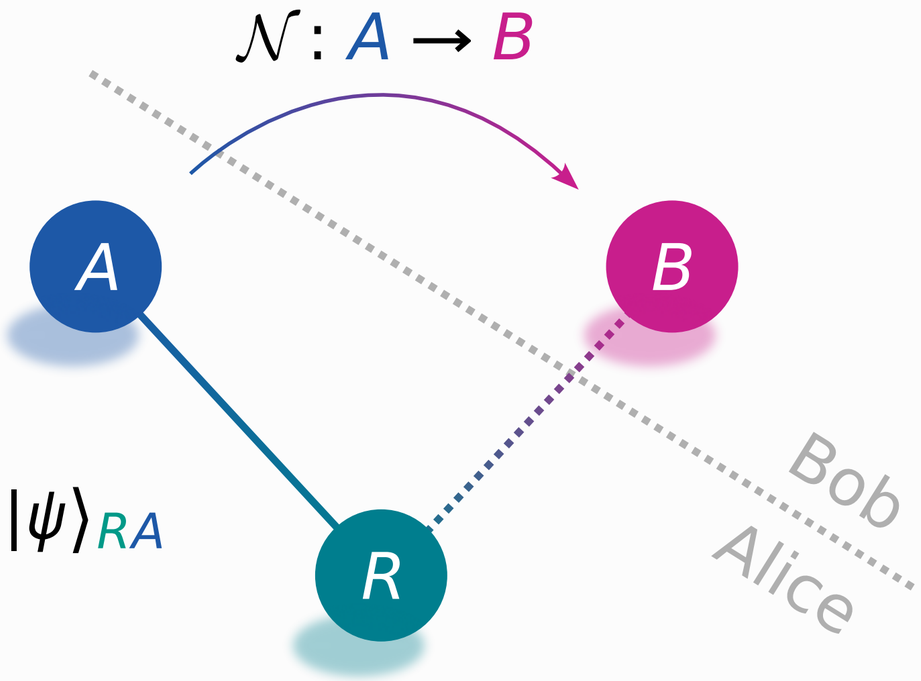
$S(B)_\sigma$ ... B vs. RE





$S(E)_\sigma$ ... E vs. RB

Quantifying quantum correlations



Coherent information

$$I_c(\mathcal{N}, \psi) = S(B)_\sigma - S(E)_\sigma$$

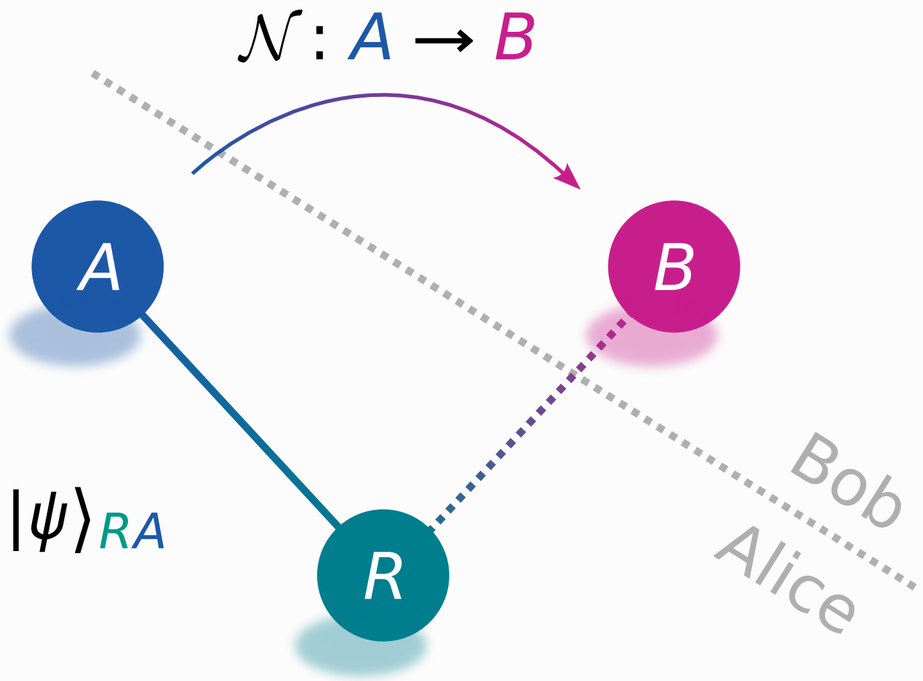
Channel coherent information

Choose best initial state ψ_{RA} between **reference** and **channel** qubit:

$$I_c(\mathcal{N}) = \max_\psi I_c(\mathcal{N}, \psi)$$

Quantifying quantum correlations

Channel coherent information $I_c(\mathcal{N}) = \max_{\psi} I_c(\mathcal{N}, \psi)$



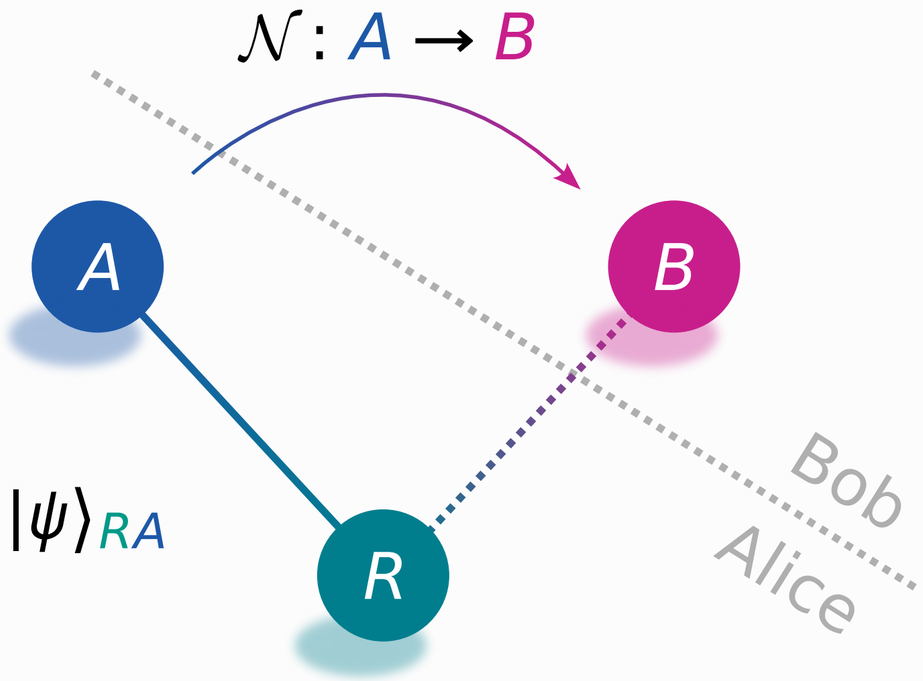
Quantum capacity theorem:

$I_c(\mathcal{N})$ many ebits per channel use can be recovered with fidelity ≈ 1 in the limit $n \to \infty$.

Lloyd '97, Shor '02, Devetak '05

Quantifying quantum correlations

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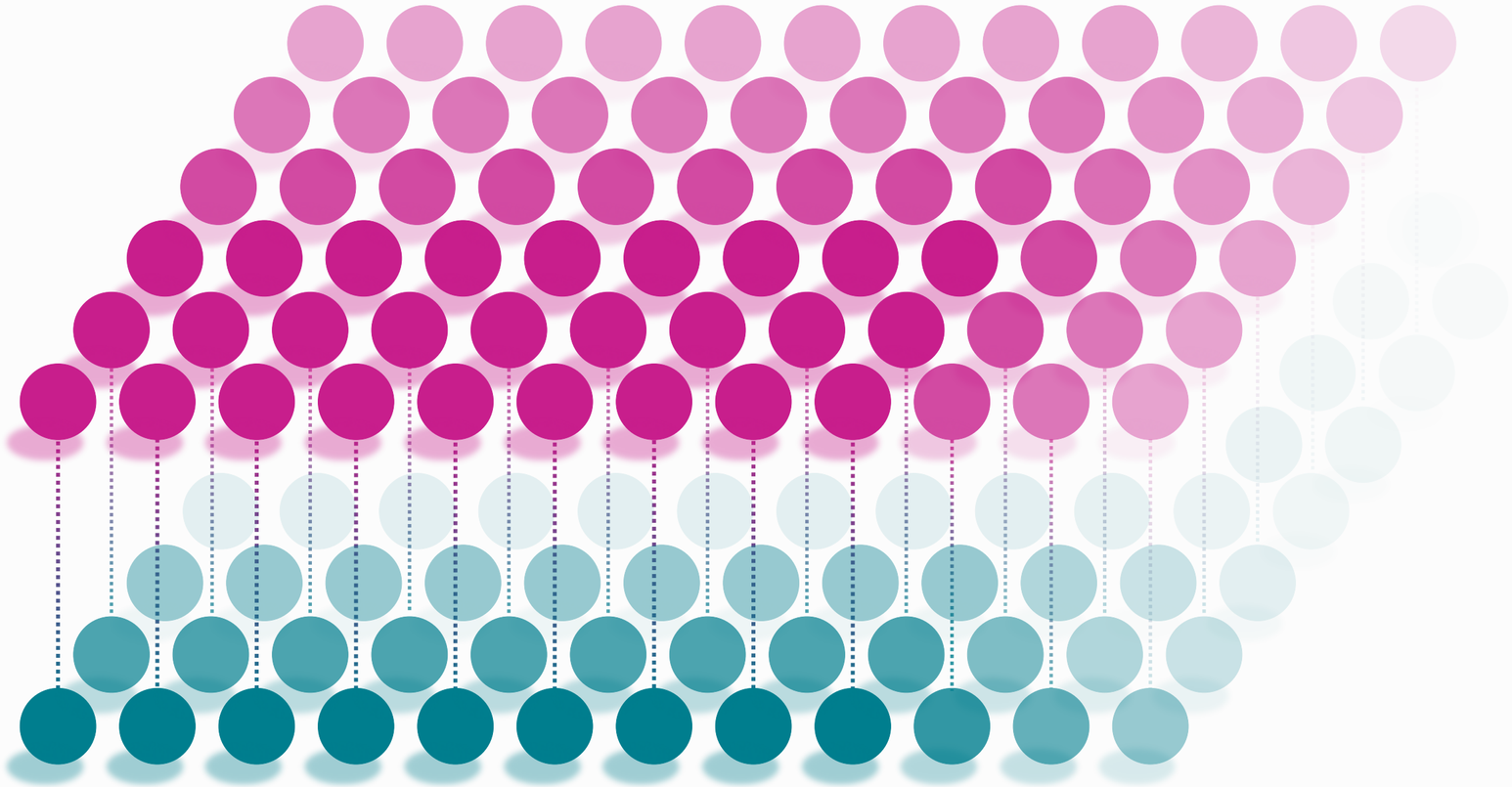
Lloyd '97, Shor '02, Devetak '05

Clearest proof:

Hayden et al., quant-ph/0702005

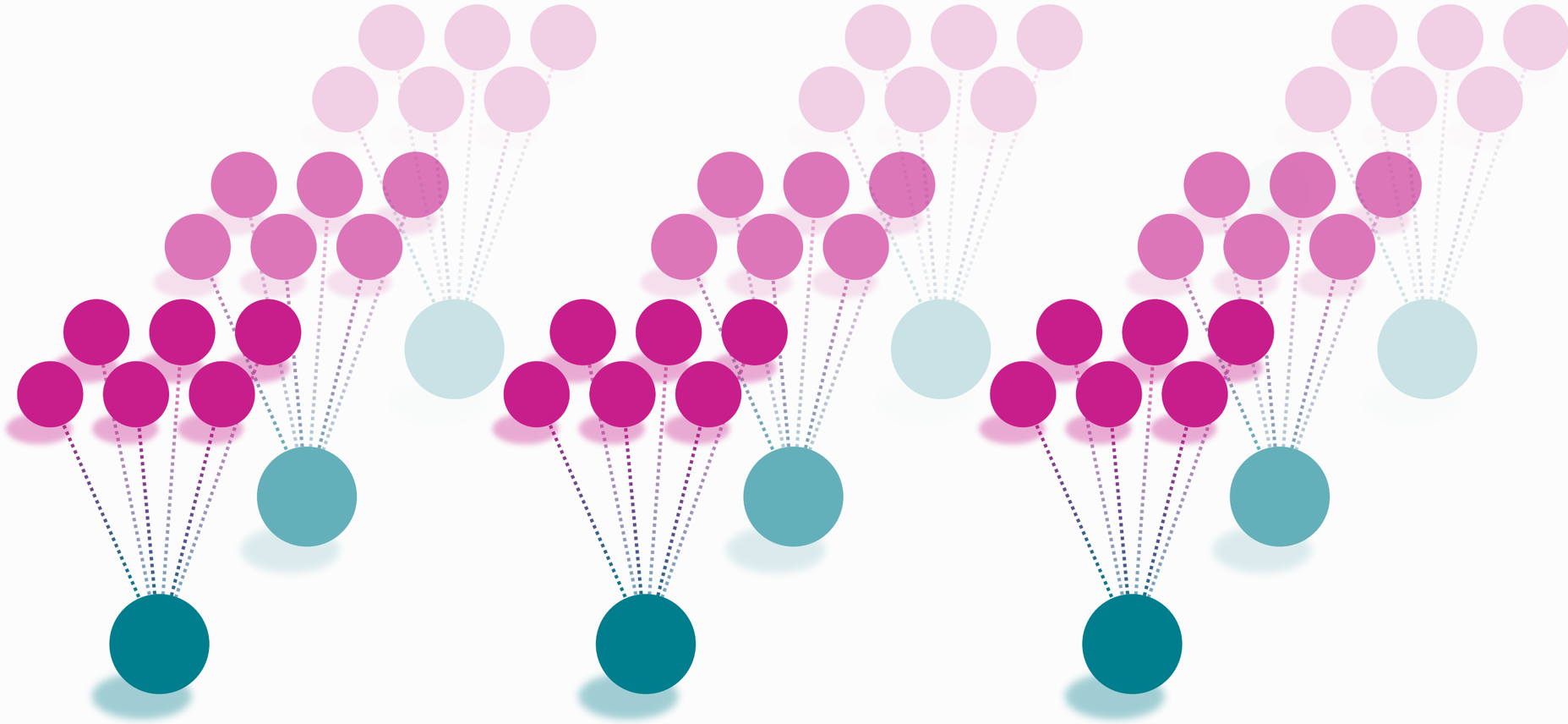
Superadditivity of coherent information

Recall: We aim to quantify surviving correlations to a reference system.



Superadditivity of coherent information

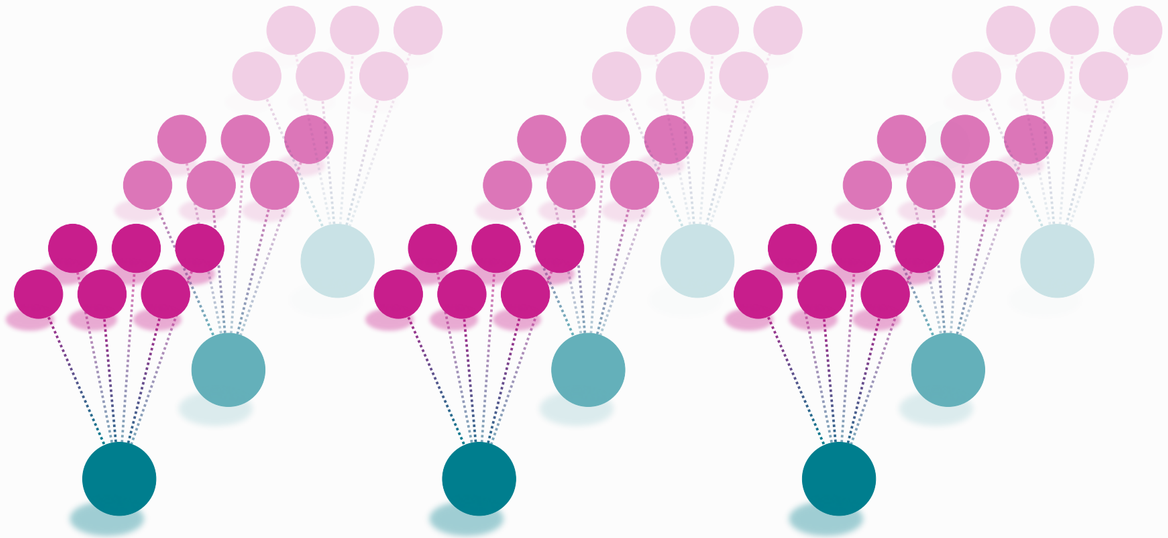
Idea: Why not use code blocks consisting of k entangled qubits?

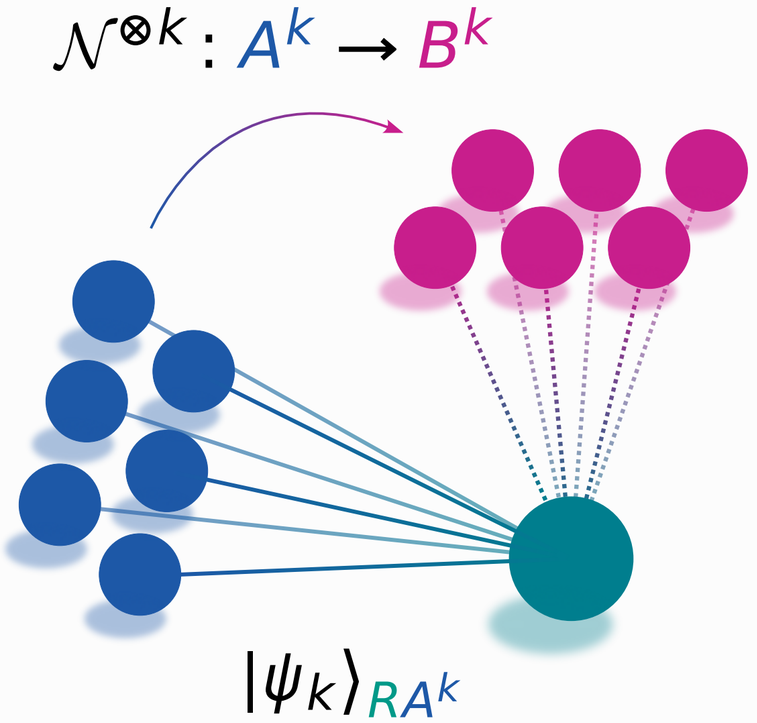


Superadditivity of coherent information

This creates blocks of k qubits in a state $|\psi_k\rangle_{RA^k}$

with average coherent information $\frac{1}{k} I_c(\mathcal{N}^{\otimes k}, \psi_k)$.



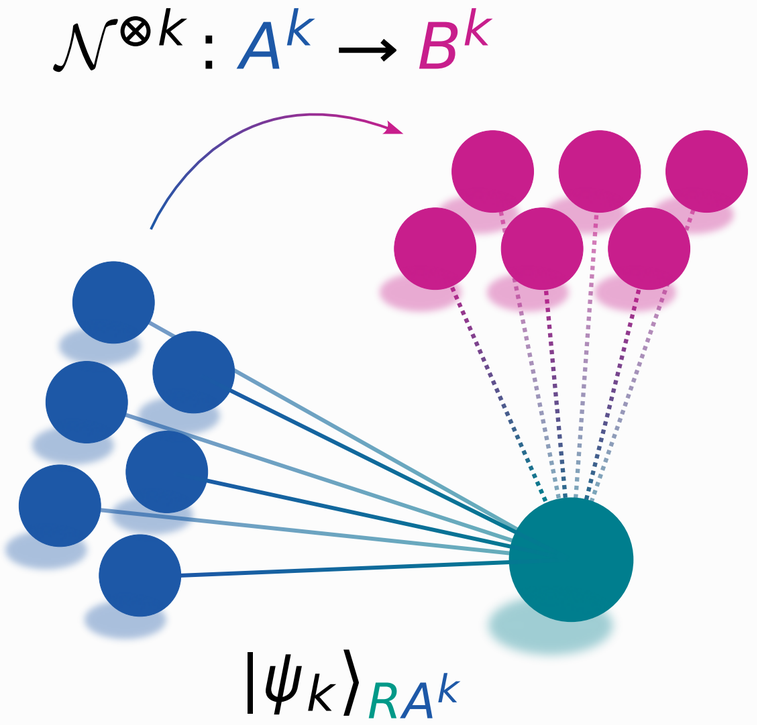


Superadditivity of coherent information

With $n = m \cdot k$ system qubits and taking $m \to \infty$,

we can recover $\frac{1}{k} I_c(\mathcal{N}^{\otimes k})$ ebits per channel use

with fidelity ≈ 1.



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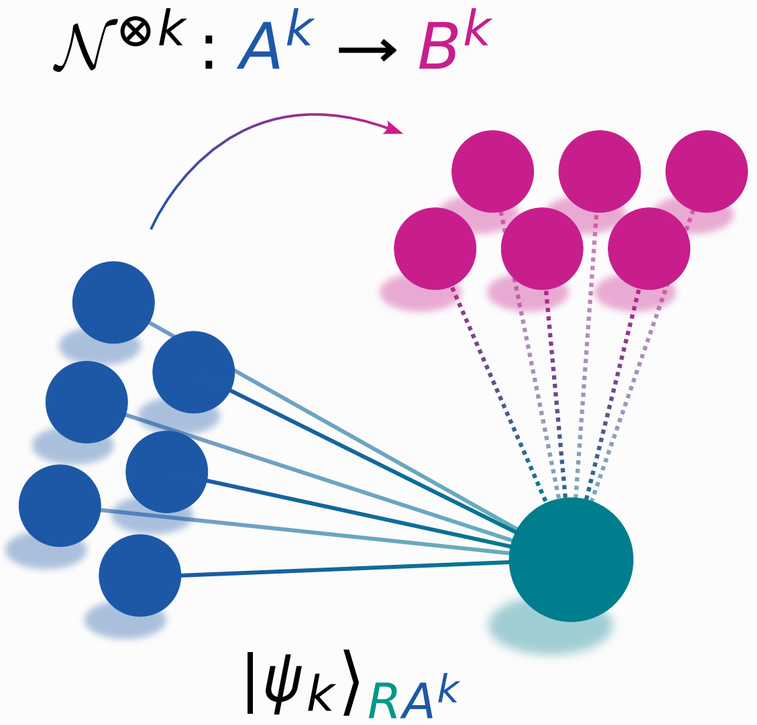
with fidelity ≈ 1.

Advantage:

Possible superadditivity of

coherent information,

$$\frac{1}{k} I_c(\mathcal{N}^{\otimes k}) > I_c(\mathcal{N}).$$



Shor & Smolin '96, DiVincenzo et al. '98, Smith & Smolin '07,...

Quantum capacity

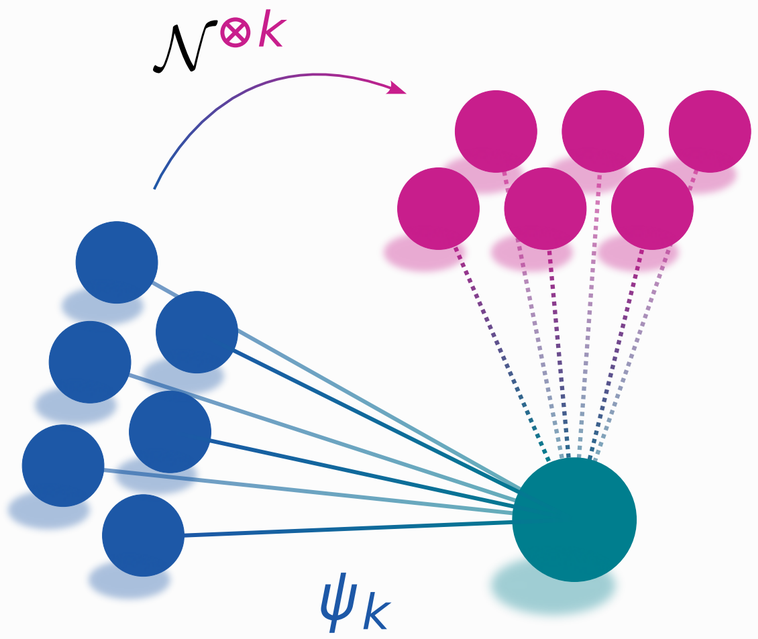
Quantum capacity $Q(\mathcal{N})$: Maximum rate at which ebits can be recovered.

Entropic formula: $$Q(\mathcal{N}) = \sup_{k \in \mathbb{N}} \frac{1}{k} I_c(\mathcal{N}^{\otimes k})$$

Schumacher, Nielsen, Barnum, Knill, Lloyd, Shor, Devetak 1996-2005

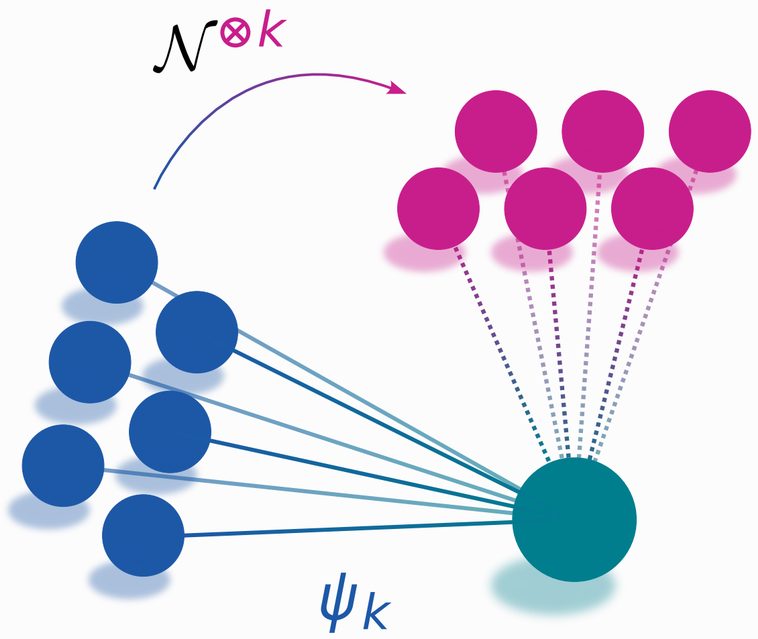
Quantum capacity

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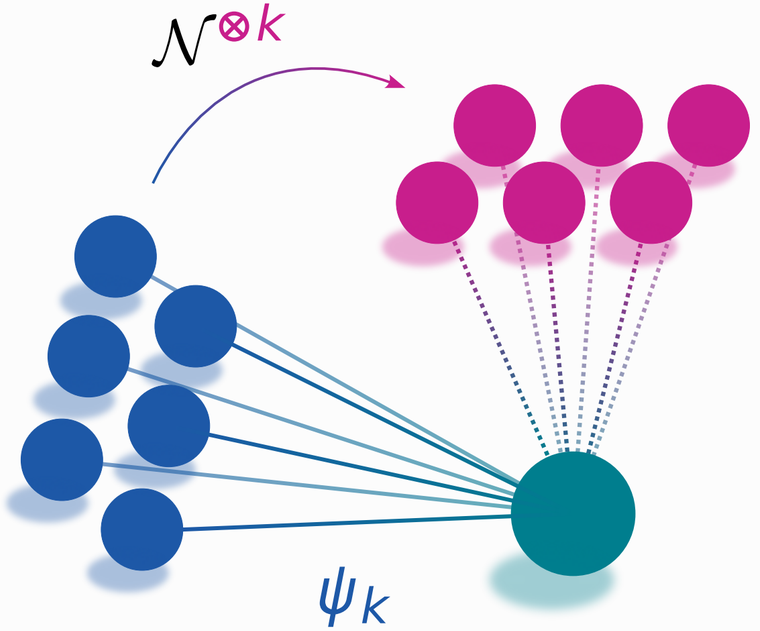
Many problems:

Unboundedness, non-convexity, multipartite entanglement, ...

Lower bounds on quantum capacity

$$Q(\mathcal{N}) = \sup_{k \in \mathbb{N}} \max_{\psi_k} \frac{1}{k} I_c(\mathcal{N}^{\otimes k}, \psi_k)$$

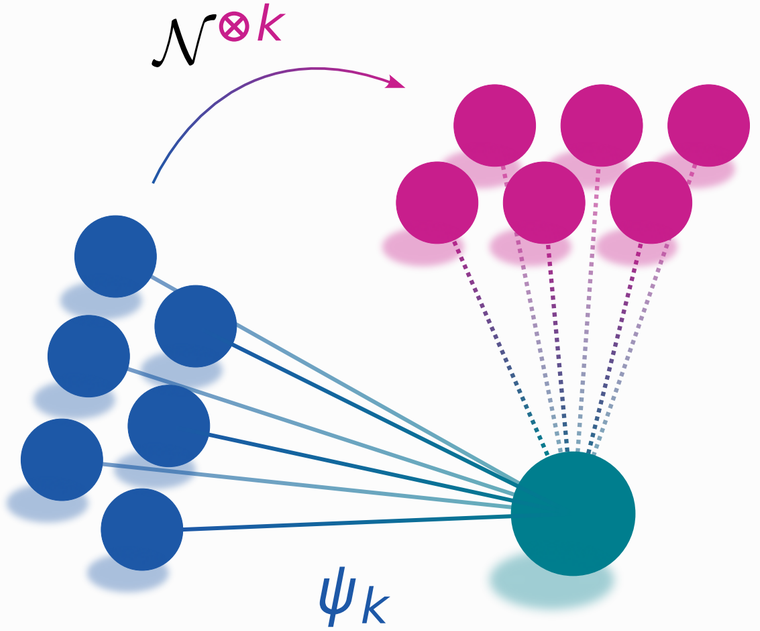
This work: Focus on lower bounds



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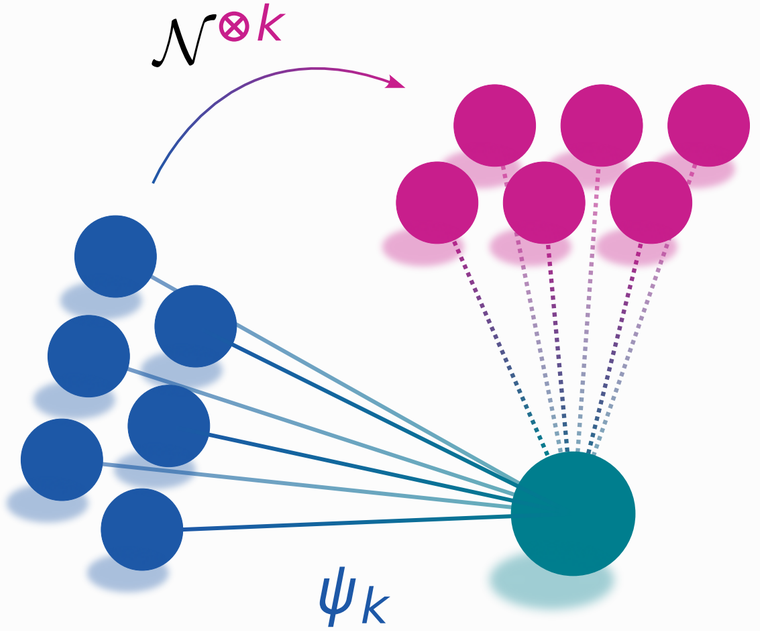
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How to efficiently compute coherent information for larger k?

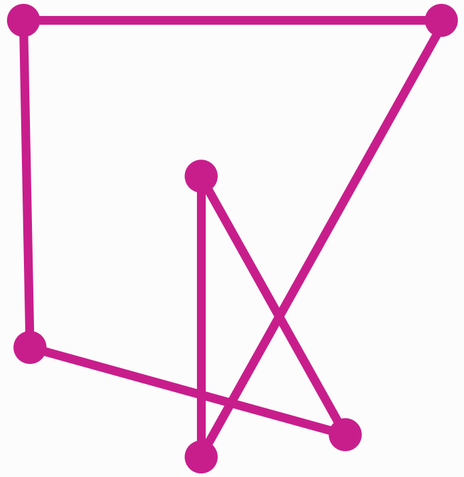
Talk outline

- Quantum information 101
- Coherent information and quantum capacity
- Symmetries in quantum information theory
- Efficient algorithms to compute entropies of symmetric states
- Quantum capacity thresholds

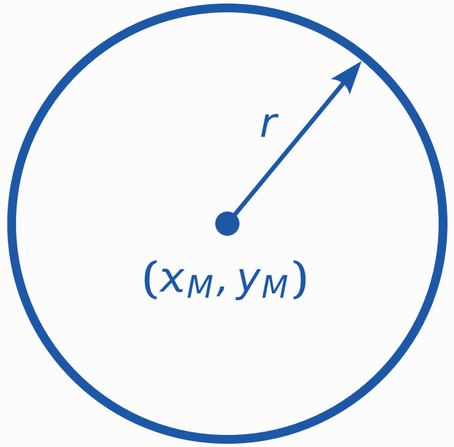
Symmetries in mathematical physics

Symmetry eliminates free parameters:

"Do something and nothing happens."



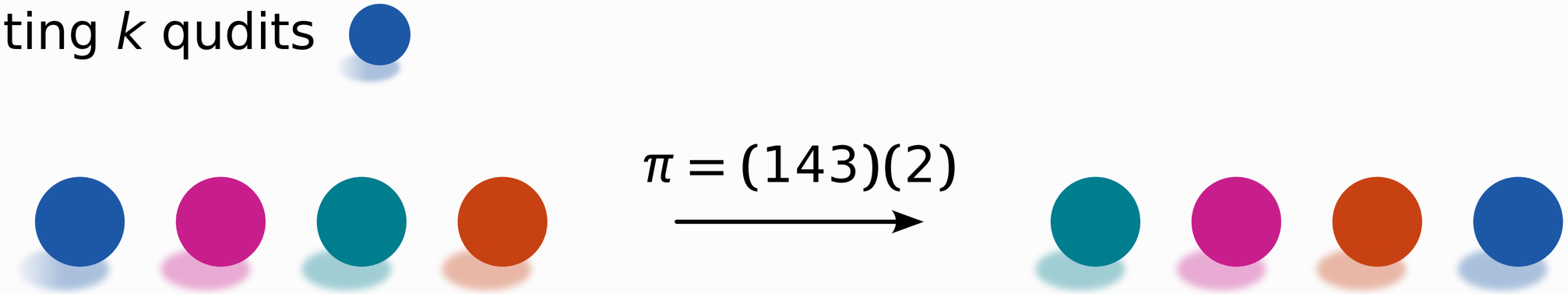
no symmetries,
complicated description



rotational symmetry,
described by three parameters

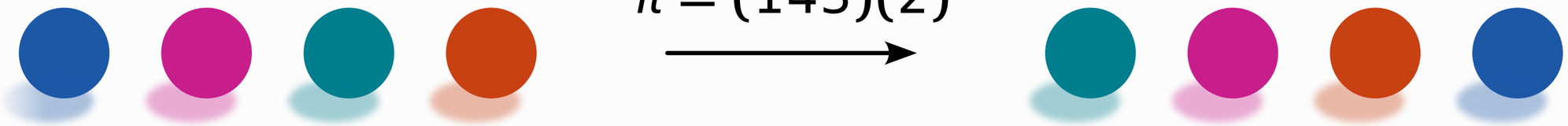
Symmetries in quantum information

Permuting k qudits

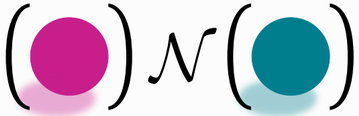


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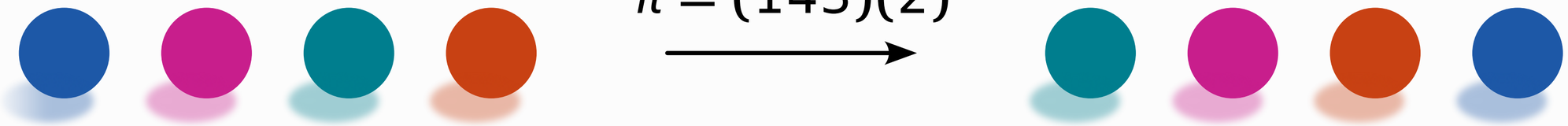


This action commutes with an i.i.d. channel $\mathcal{N}^{\otimes k}$:

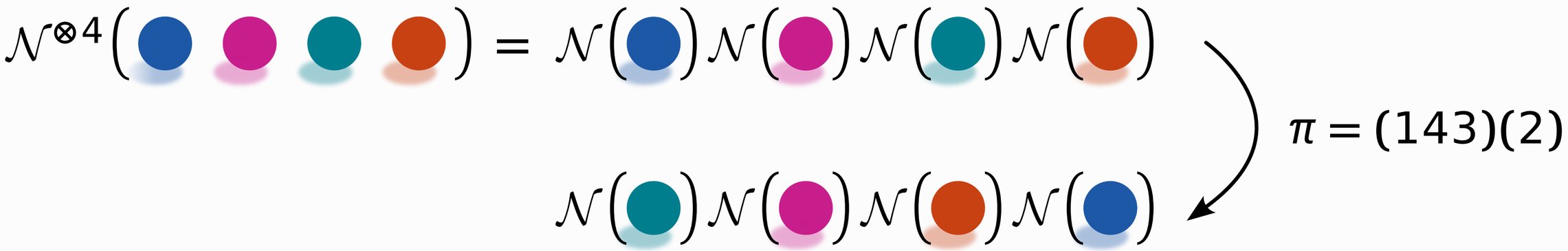
$\mathcal{N}^{\otimes 4}$  $= \mathcal{N}$  $\mathcal{N}$ $\mathcal{N}$  $\mathcal{N}$

Symmetries in quantum information

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Decomposing representations

Permuting qudits is a **representation** of the symmetric group S_k on the Hilbert space $(\mathbb{C}^d)^{\otimes k}$.

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We can decompose this representation into “building blocks” called **irreducible representations** (irreps) S_λ:

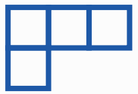
$$(\mathbb{C}^d)^{\otimes k} \cong \bigoplus_\lambda S_\lambda \oplus \cdots \oplus S_\lambda$$

$$\cong \bigoplus_\lambda S_\lambda \otimes M_\lambda$$

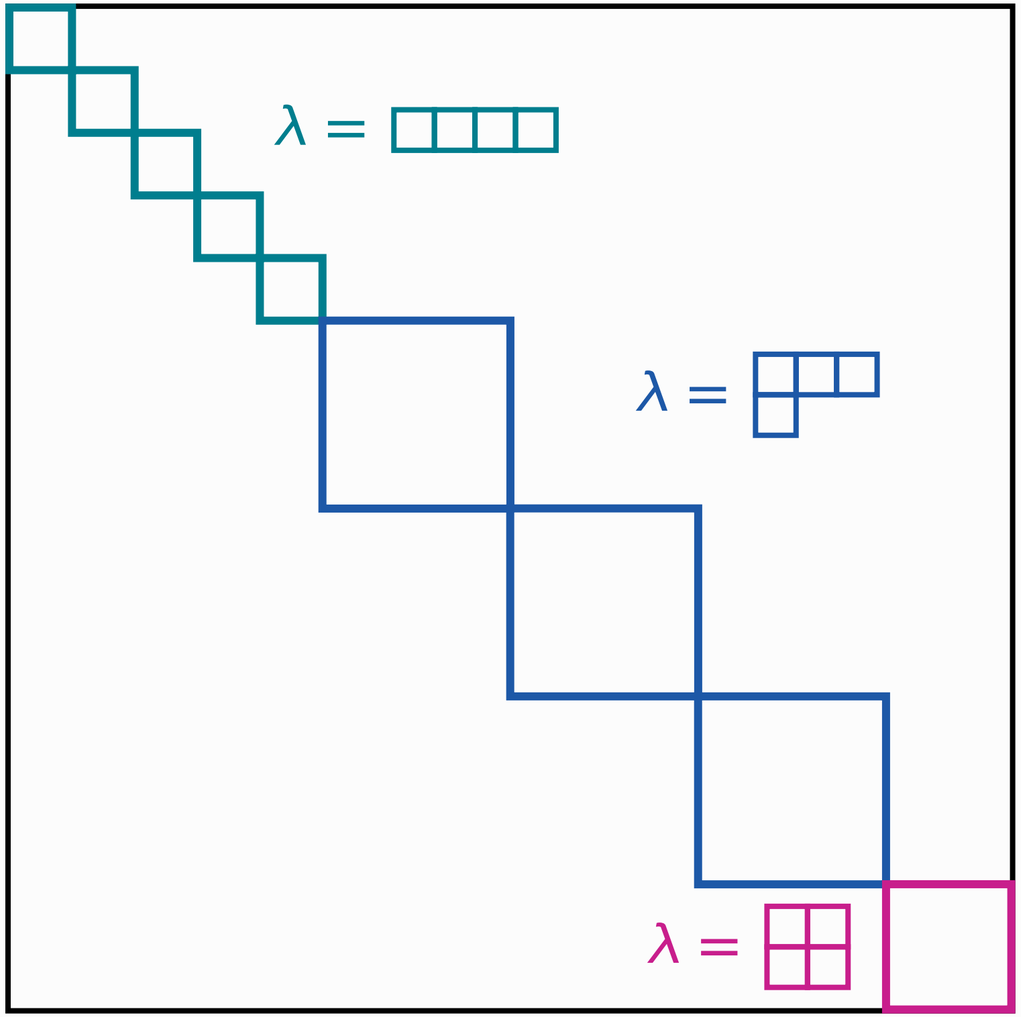
λ labels inequivalent **irreps** S_λ, and M_λ is a **multiplicity space**.

Decomposing representations

$$(\mathbb{C}^d)^{\otimes k} \cong \bigoplus_{\lambda} S_\lambda \otimes M_\lambda$$

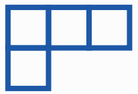
λ			
$\dim S_\lambda$	1	3	2
$\dim M_\lambda$	5	3	1

Example: $d = 2, k = 4$



Decomposing representations

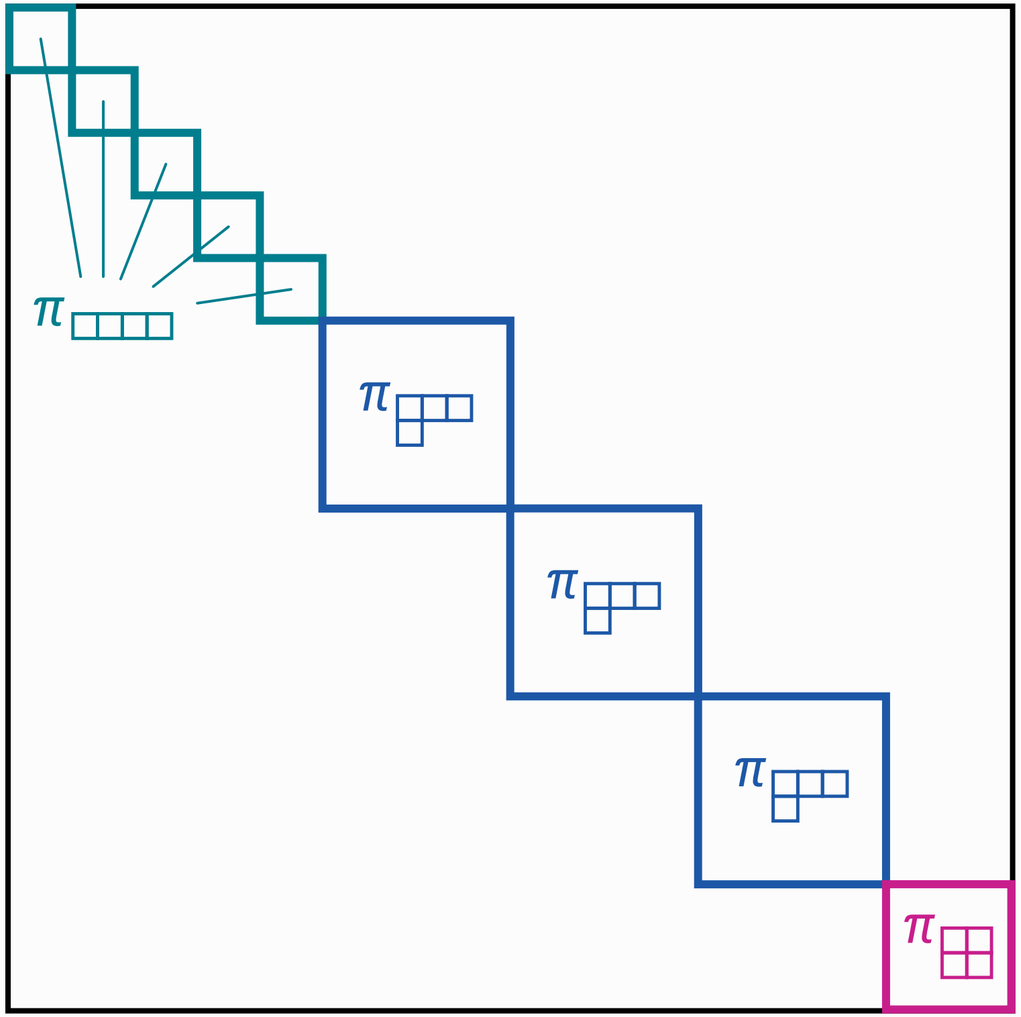
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Example: $d = 2, k = 4$

Permutations only act on the S_λ:

$$\pi \cong \bigoplus_{\lambda} \pi_\lambda \otimes \mathbb{1}_{M_\lambda}$$



Schur's Lemma

What do invariant states look like?

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Schur's Lemma: Let π_λ and π_μ be two irreps of a group G, and X be a linear map satisfying

$$X\pi_\lambda(g) = \pi_\mu(g)X \text{ for all } g \in G.$$

$$\text{Then } X = \begin{cases} 0 & \text{if } \lambda \ncong \mu; \\ c\mathbb{1}_{S_\lambda} & \text{if } \lambda \cong \mu. \end{cases}$$

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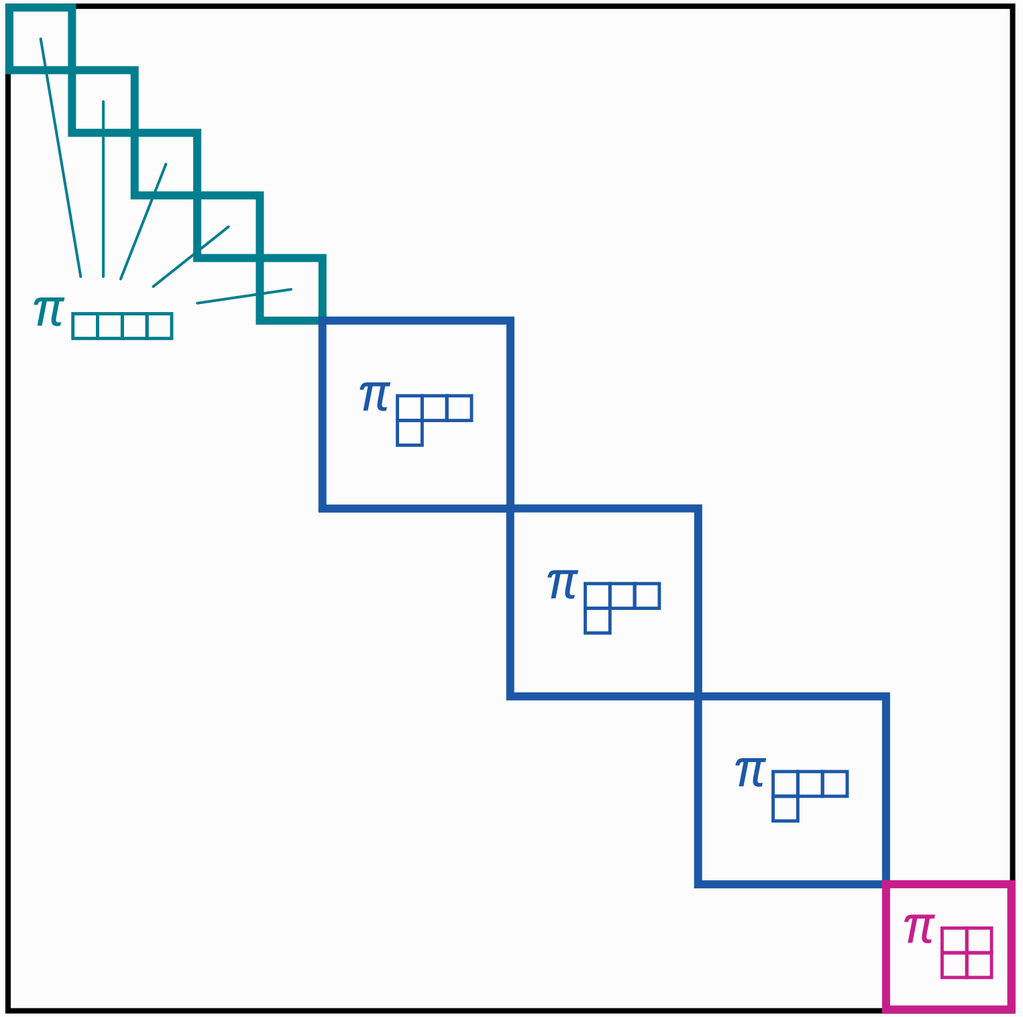
Extremely simple to prove, **extremely useful** in applications!

Invariant states

Space decomposition

$$(\mathbb{C}^d)^{\otimes k} \cong \bigoplus_{\lambda} S_\lambda \otimes M_\lambda$$

$$S_k \ni \pi \cong \bigoplus_{\lambda} \pi_\lambda \otimes \mathbb{1}_{M_\lambda}$$



Invariant states

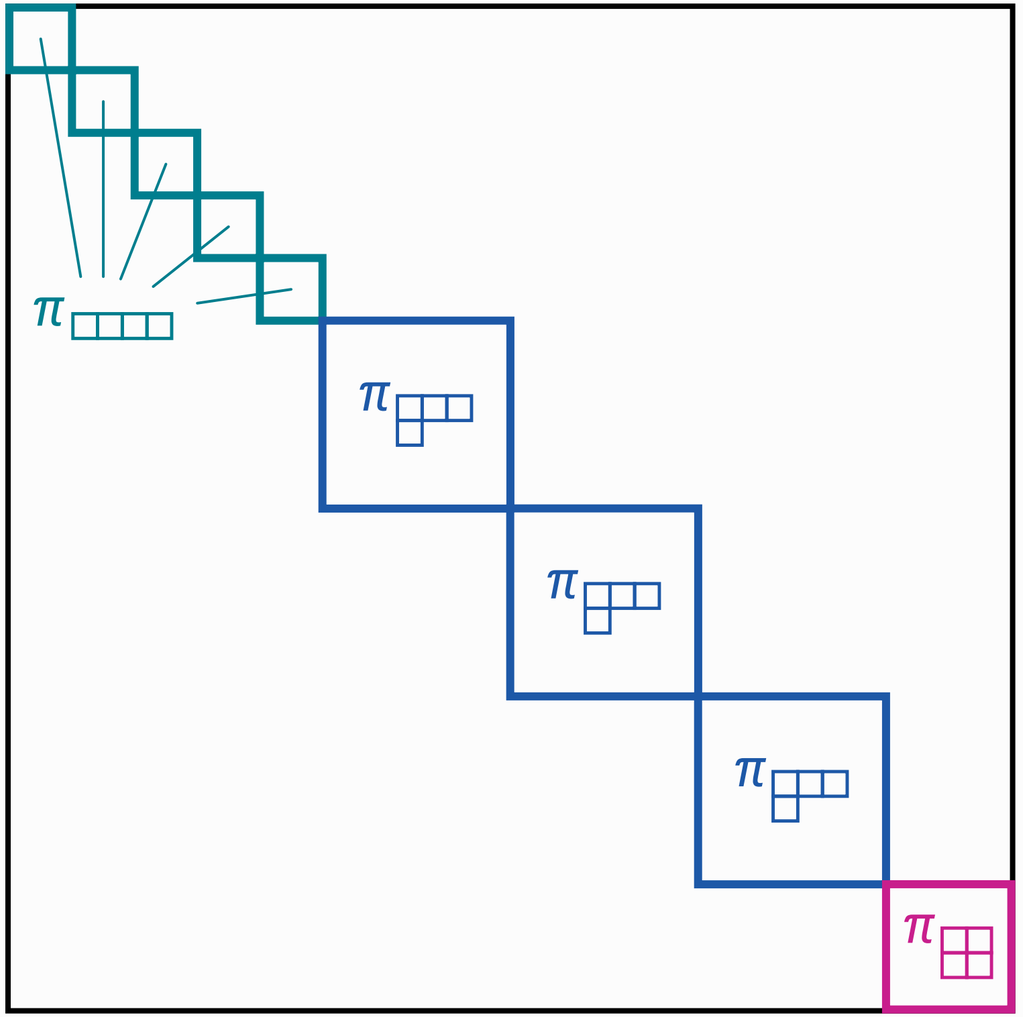
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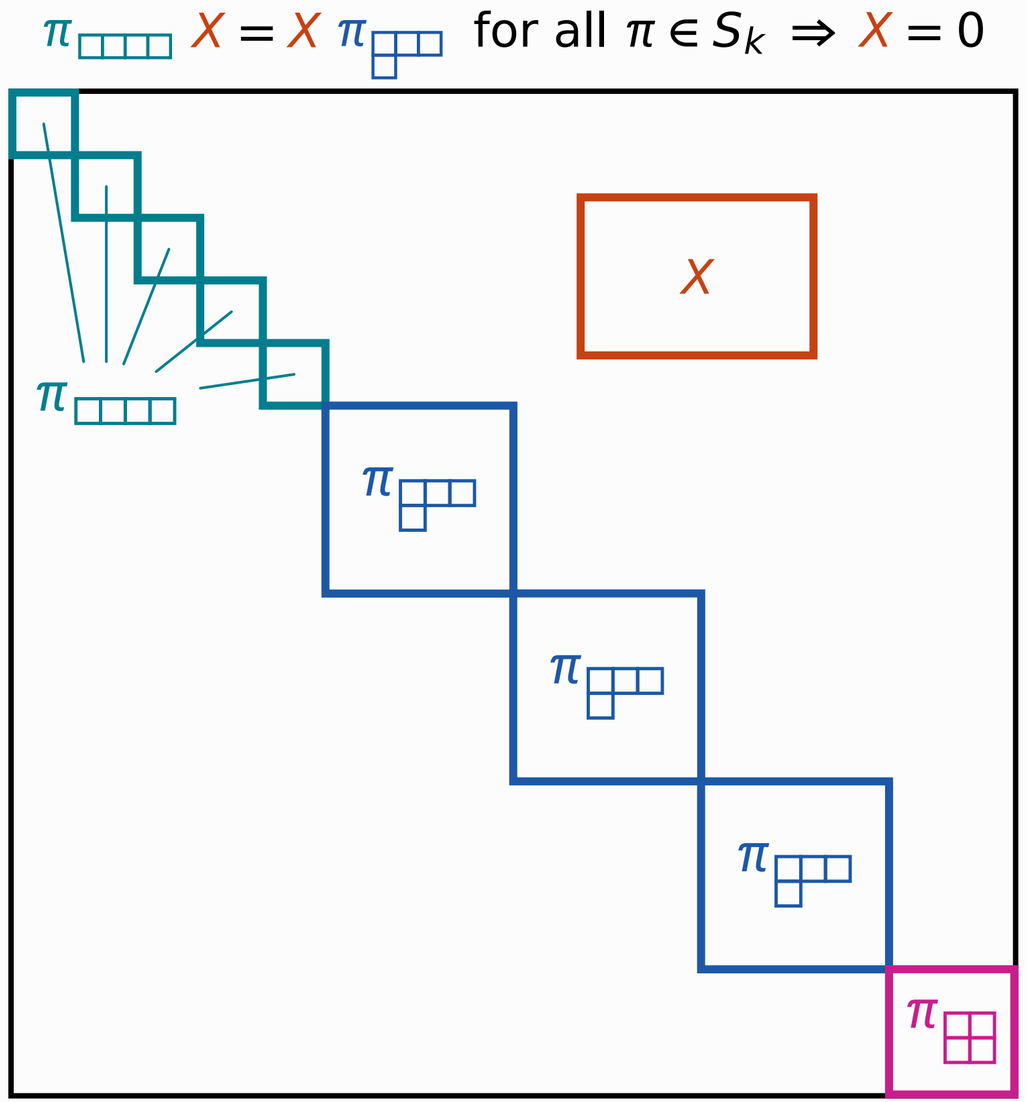
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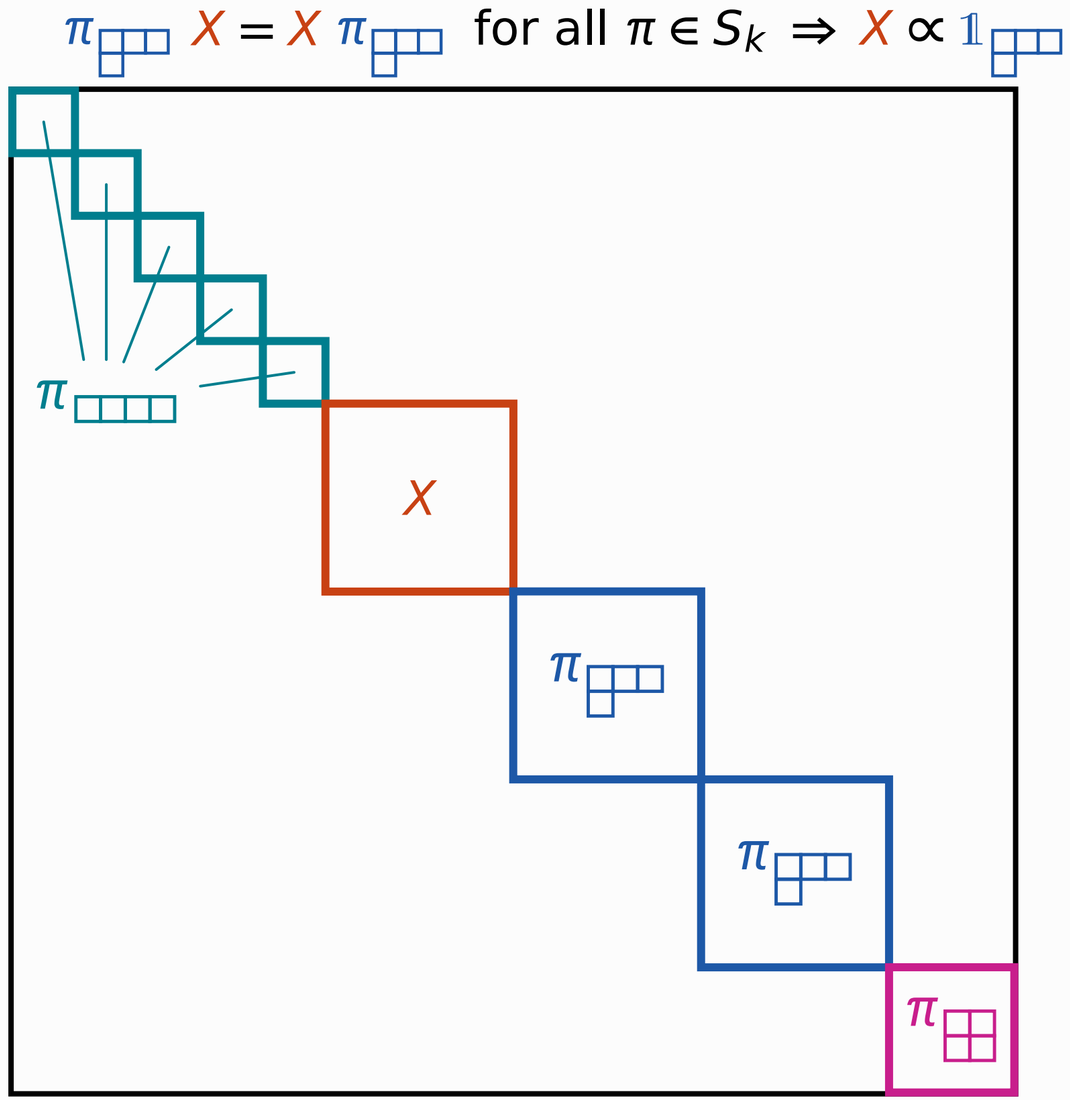
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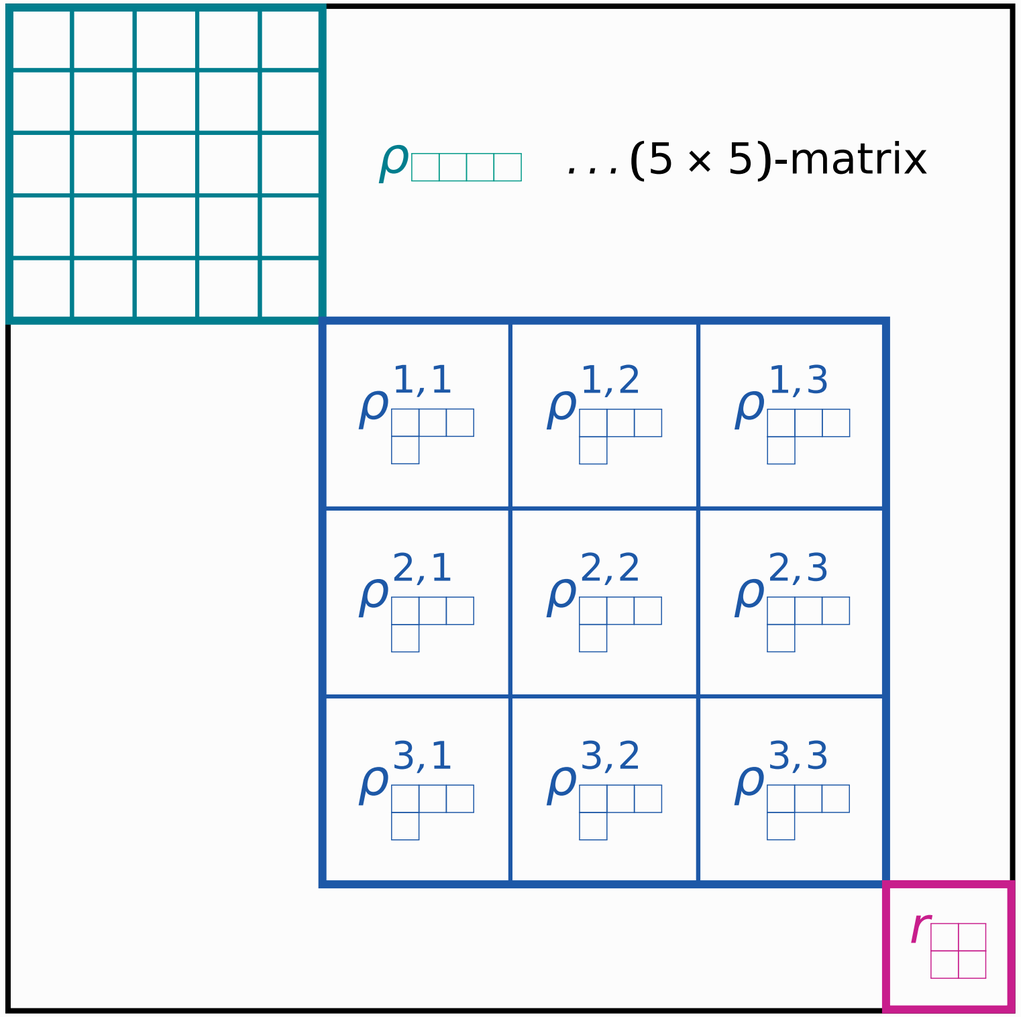
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Invariant states

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How can we determine the effective states ρ_λ?

Invariant states

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Schur-Weyl duality

The multiplicity spaces M_λ in the decomposition

$$(\mathbb{C}^d)^{\otimes k} \cong \bigoplus_{\lambda} S_\lambda \otimes M_\lambda$$

are actually the irreps of the representation $\mathrm{GL}(d) \ni g \mapsto g^{\otimes k}$
of the general linear group $\mathrm{GL}(d)$.

Invariant states

Recall:

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This lets us easily describe states of the form $\rho_{(k)} = \sum_i x_i \rho_i^{\otimes k}$, Molev '06

as well as more general permutation-invariant states.

Invariant states

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Well-known example: Repetition code

$$\rho_{(k)} = \tfrac{1}{2}\left(|0\rangle\langle 0|^{\otimes k} + |1\rangle\langle 1|^{\otimes k}\right)$$

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This code gives **superadditive coherent information** for Pauli channels

$$\rho \longmapsto (1-p)\rho + p\left[q_1 X\rho X + q_2 Y\rho Y + q_3 Z\rho Z\right]$$

Shor & Smolin '96; Smith & Smolin '07; Bausch & **FL** '21

Main information-theoretic result

The symmetric ansatz gives an **efficient way of computing entropies**.

Well-known property of von Neumann entropy $S(\rho) = -\operatorname{tr} \rho \log \rho$:

$$\text{If } \rho = \bigoplus_i p_i \rho_i, \text{ then } S(\rho) = H(\mathbf{p}) + \sum_i p_i S(\rho_i).$$

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We apply this to channel outputs of symmetric states such as

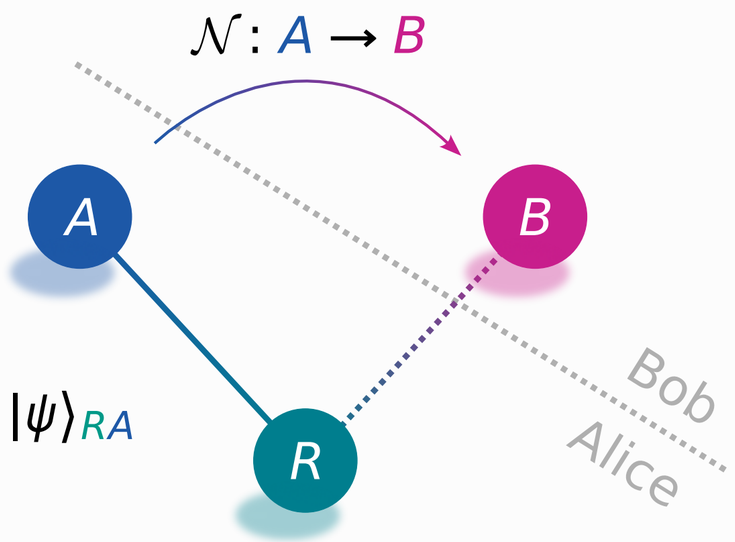
$$\sigma_{(k)} = \sum_i x_i \mathcal{N}(\rho_i)^{\otimes k} \cong \bigoplus_\lambda p_\lambda \frac{1}{\dim S_\lambda} \mathbb{1}_{S_\lambda} \otimes \sum_i x_i \mathcal{N}(\rho_i)^{(\lambda)}.$$

Example: Coherent information

For states of the form $\rho_{(k)} = \sum_i x_i \rho_i^{\otimes k} \cong \bigoplus_\lambda p_\lambda \frac{1}{\dim s_\lambda} \mathbb{1}_{s_\lambda} \otimes \sum_i x_i \rho_i^{(\lambda)}$,

the coherent information $I_c(\mathcal{N}^{\otimes k}, \rho_{(k)})$ assumes the form

$$I_c(\mathcal{N}^{\otimes k}, \rho_{(k)}) = \sum_\lambda p_\lambda f\left(\sum_i x_i \mathcal{N}(\rho_i)^{(\lambda)}\right).$$

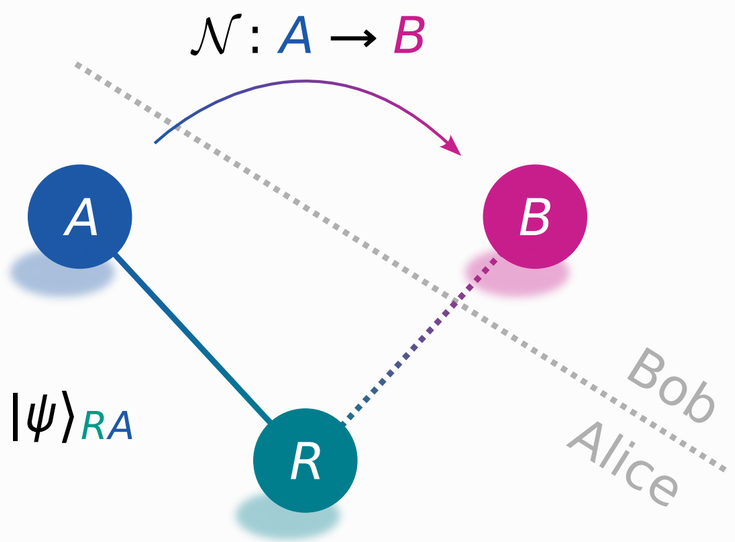


$I_c(\mathcal{N}, \psi) = S(B)_\sigma - S(RB)_\sigma$

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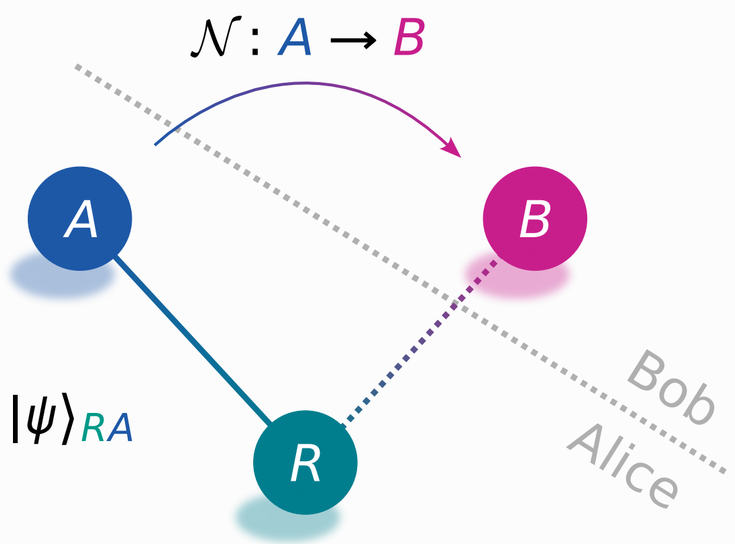
polynomially many terms

polynomially sized operators

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polynomially many terms

polynomially sized operators

$\mathcal{O}(\text{poly}(n))$ vs. $\mathcal{O}(\exp(n))$:

up to $n \approx 100$ for qubit channels!

Talk outline

- Quantum information 101
- Coherent information and quantum capacity
- Symmetries in quantum information theory
- Efficient algorithms to compute entropies of symmetric states
- Quantum capacity thresholds

Application: Finding thresholds

Goal: Find lower bounds on the quantum capacity of channels such as

$$\mathcal{N}_p : \sigma \longmapsto (1-p)\sigma + p\left[q_1 X\sigma X + q_2 Y\sigma Y + q_3 Z\sigma Z\right]$$

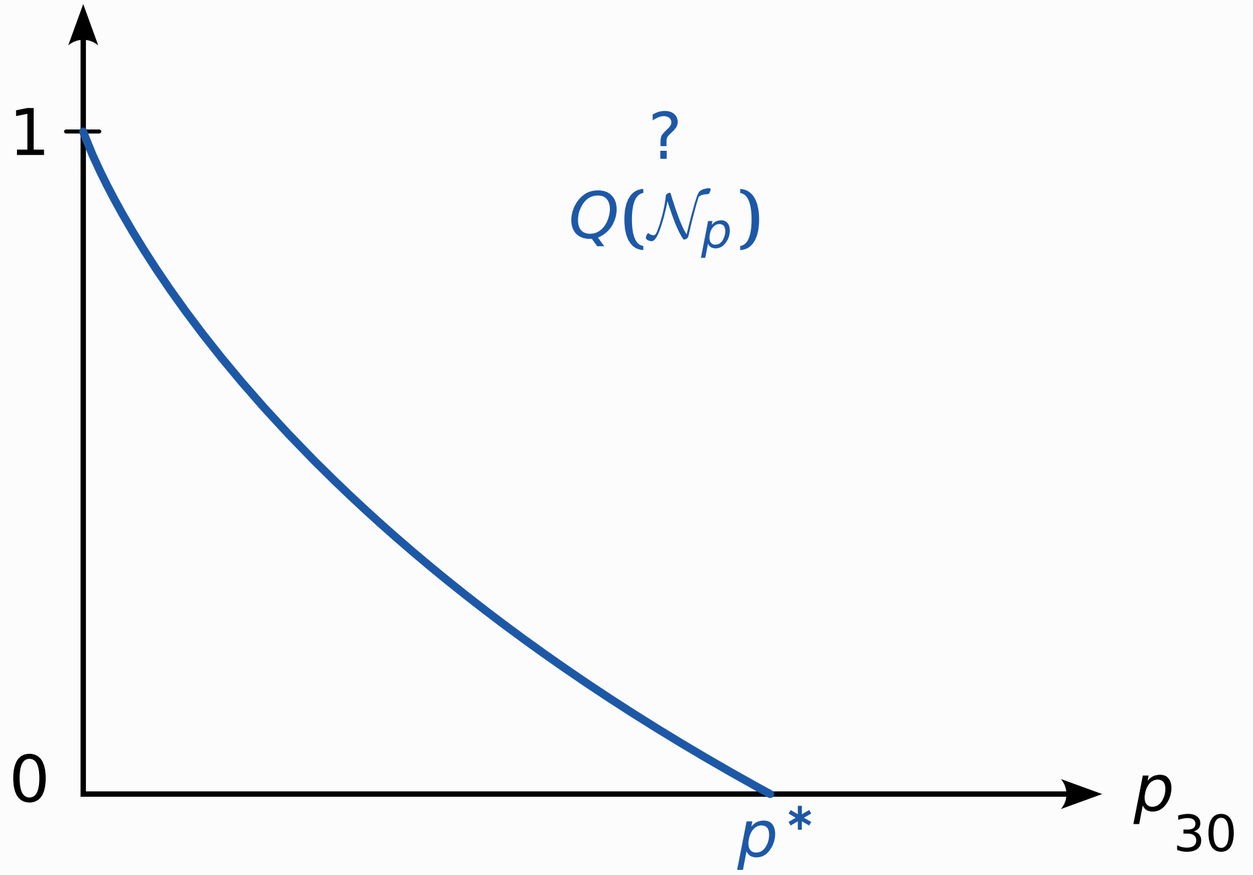
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Quantum capacity threshold:

$$p^* = \sup\{p : Q(\mathcal{N}_p) > 0\}$$



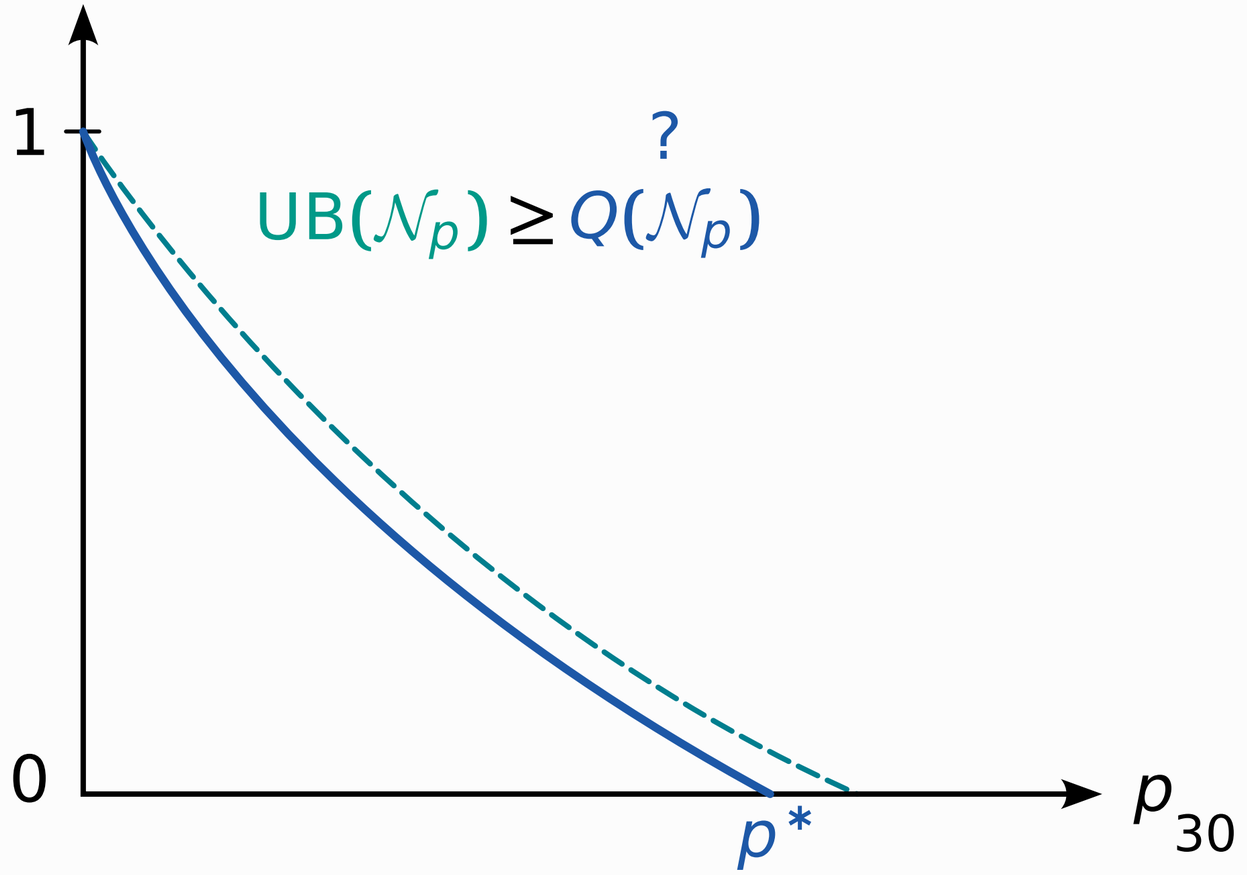
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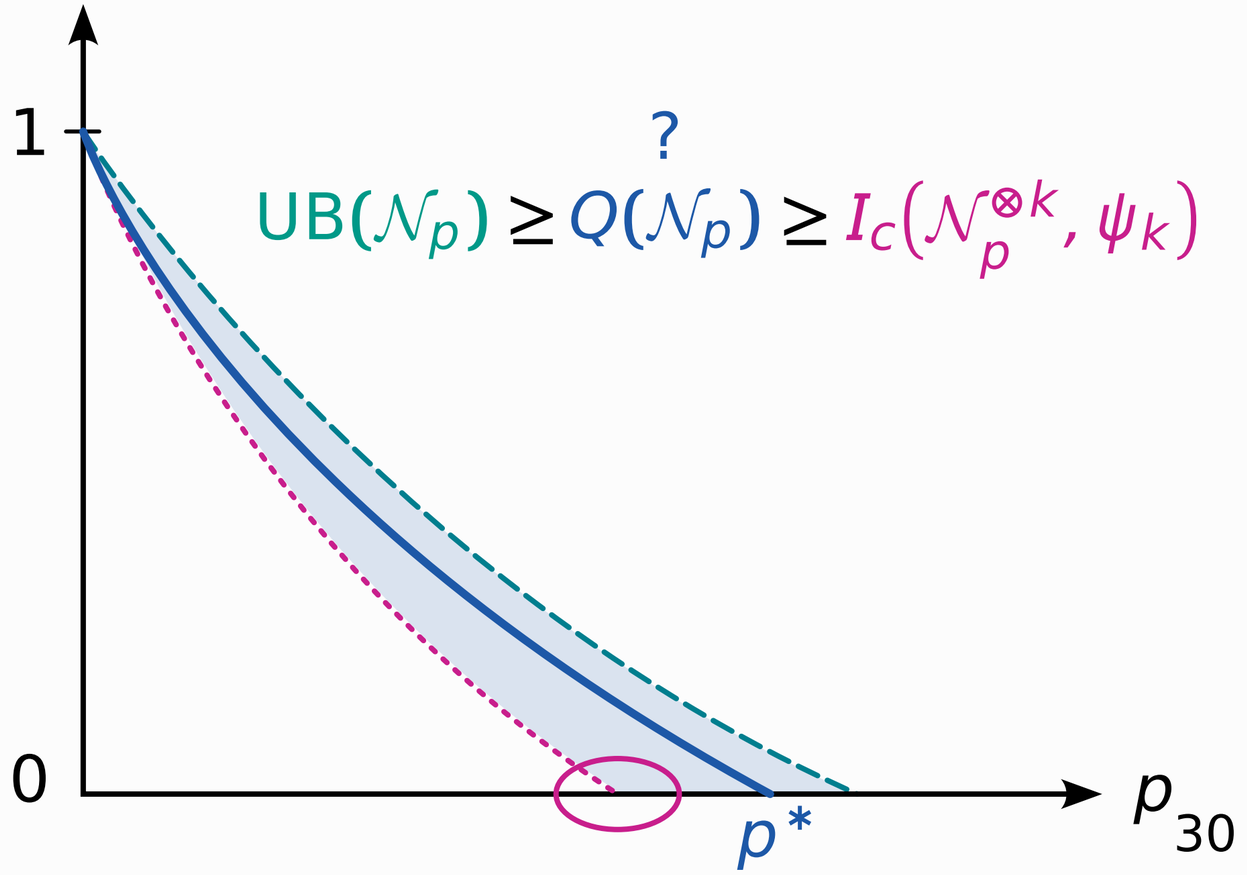
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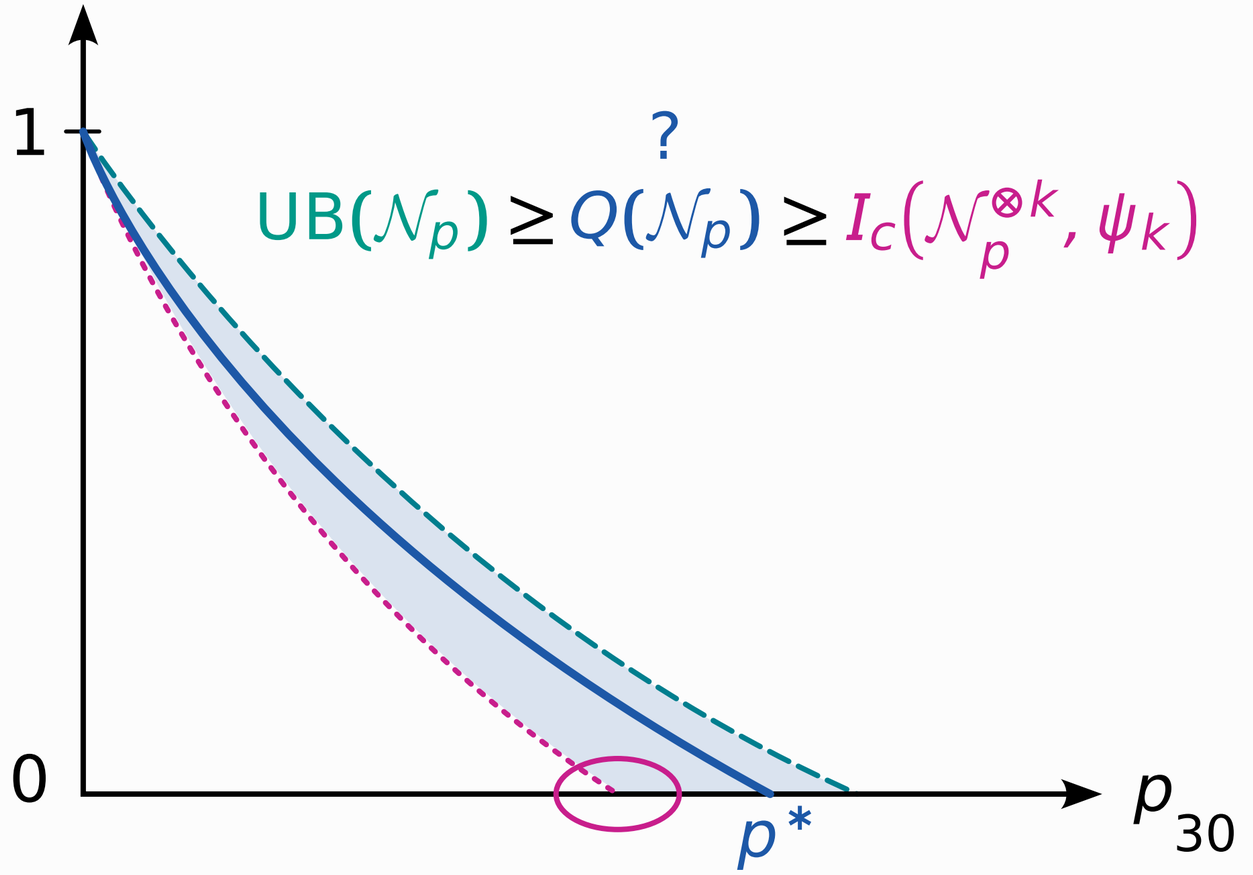
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$$\implies p^* \geq p_0$$



Thresholds of important Pauli channels

Depolarizing channel $\rho \mapsto (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$

Upper bound on threshold: $p^* \leq 1/4$

Previous lower bound: $p^* \gtrsim 0.1913$ ⟶ improved to $p^* \gtrsim 0.1989$

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BB84-channel $\rho \mapsto (1-p)^2\rho + p(1-p)\,X\rho X + p^2\,Y\rho Y + p(1-p)\,Z\rho Z$

Upper bound on threshold: $p^* \leq 0.1464$

Previous lower bound: $p^* \gtrsim 0.1128$ ⟶ improved to $p^* \gtrsim 0.1184$

[Bhalerao & FL '25, Agarwal et al. '26, ongoing work]

Further analytic results

Some states achieving improved thresholds have an interesting structure:

$$\rho_{(n)} = \frac{1}{2}\left(|\psi_1\rangle\langle\psi_1|^{\otimes n} + |\psi_2\rangle\langle\psi_2|^{\otimes n}\right) \quad \text{with} \quad \langle\psi_1|\psi_2\rangle \neq 0.$$

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The works [Bhalerao & FL '25] and [Agarwal et al. '26] carry out a representation-theoretic analysis for why these and other permutation-invariant states are useful ansätze in entropic optimization problems.

Future work

Take-away: Permutation invariance is not just a mathematical device that improves scaling, but actually a good ansatz to describe multipartite states and their interaction with noise.

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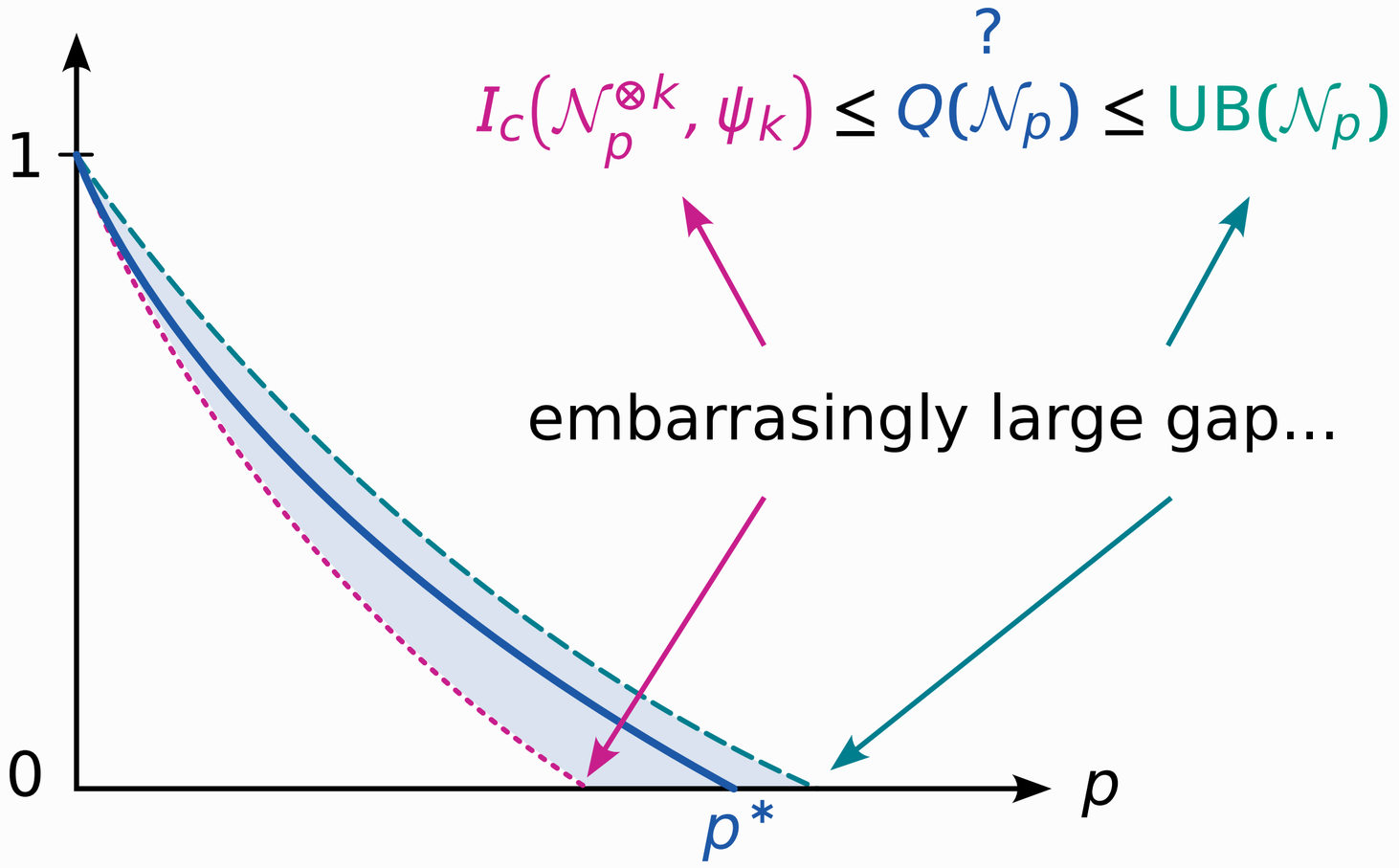
Major open problem in QIT: Find explicit channel exhibiting this!

Related to non-additivity of **minimum output entropy**:

$$S_{\min}(\mathcal{N}) = \min_{\psi} S(\mathcal{N}(\psi))$$

Long-term future work

Determine the quantum capacity of Pauli channels!



Links to paper and code

[Bhalerao & FL '25]

Code on GitHub

See also
[Agarwal et al. '26] at

arXiv:2605.09138

Thanks for your attention!